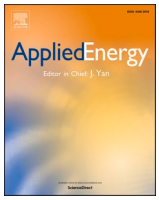
Applied Energy 414 (2026) 127841

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy



Integrated planning of wind–solar–storage–grid systems for private-wire supplied data centers

Feng Guo [a,b], João Soares [c], Ling Ma [a,b,*], Jie Song [d], Zhiyong Tian [e,f], Xinyu Chen [g], Zita Vale [c]

[a] National Center of Technology Innovation for Digital Construction, Huazhong University of Science and Technology, Wuhan, 430074, China
[b] School of Civil and Hydraulic Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China
[c] Research Group on Intelligent Engineering and Computing for Advancement Innovation and Development (GECAD), Intelligent Systems Associate Laboratory (LASI), School of Engineering - ISEP, Polytechnic of Porto, 4249-015 Porto, Portugal
[d] School of Advanced Manufacturing and Robotics, Peking University, Beijing, 100871, China
[e] School of Environmental Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China
[f] Hubei Key Laboratory of Multi-media Pollution Cooperative Control in Yangtze Basin, School of Environmental Science & Engineering, Huazhong University of Science and Technology, Wuhan, 430074, China
[g] School of Electrical and Electronic Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

HIGHLIGHTS

- Develops a private-wire co-planning MILP with PCC and hourly deliverability constraints.
- Reconstructs an 8760-h climate-coupled load for a standardised 25,000-rack campus.
- Simulates hourly wind and solar PV generation from ERA5 reanalysis.
- Reveals a shift from grid-trading balancing to capital-intensive local supply at high targets.
- Quantifies per-city cost, grid purchase displacement, and implied CO_2 abatement.

ARTICLE INFO

Keywords:
Data centers
Private wire
Power usage effectiveness
Renewable energy
Mixed integer linear programming
Sustainable development

ABSTRACT

Rapid data center expansion intensifies the tension between deep power-sector decarbonisation and the need for highly reliable, continuous electricity supply. Under grid congestion and limited interconnection capacity, annual renewable claims cannot guarantee hour-by-hour deliverable clean power. We develop an integrated planning framework for private-wire-supplied data centers. This framework determines the optimal within-city siting of the data center hub and its matched wind, solar photovoltaic, and battery resources, subject to point-of-common-coupling exchange limits and dedicated-line deliverability constraints. Candidate renewable sites are identified through spatial screening; hourly generation is simulated using ERA5 reanalysis; and an 8760-h load profile is reconstructed for a standardised 25,000-rack campus using a climate-driven power usage effectiveness model. A mixed-integer linear programming model then co-optimises siting, capacities, line sizing, and grid transactions under renewable supply targets of 30%, 50%, 70%, and 95%. Using Qinghai, China, as an illustrative case, we find that hub siting depends mainly on interconnection access and coupling conditions, whereas renewable siting increasingly concentrates in the strongest resource corridors as the target tightens. Solar photovoltaics form the main scalable backbone, while wind is used selectively only in the most advantageous zones. Across candidate-city cases, total annual cost rises from 1.76 to 1.96 billion USD at 30% to 8.56–9.56 billion USD at 95%. Annual displacement of grid electricity purchases increases from about 0.76 TWh to 2.17–2.26 TWh per city, and implied emissions reductions rise from about 0.14 to 0.39–0.41 Mt. CO_2 per city. The key finding is that, at high clean-electricity shares, the main constraint is not annual renewable availability but hour-by-hour deliverability within the project boundary.

* Corresponding author at: National Center of Technology Innovation for Digital Construction, Huazhong University of Science and Technology, Wuhan 430074, China
E-mail addresses: guofeng777@hust.edu.cn (F. Guo), jan@isep.ipp.pt (J. Soares), ling_ma@hust.edu.cn (L. Ma), jie.song@pku.edu.cn (J. Song), zhiyongtian@hust.edu.cn (Z. Tian), xchen2019@hust.edu.cn (X. Chen), zav@isep.ipp.pt (Z. Vale).

https://doi.org/10.1016/j.apenergy.2026.127841
Received 7 January 2026; Received in revised form 22 March 2026; Accepted 31 March 2026
Available online 7 April 2026
0306-2619/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

Abbreviations

Abbreviation	**Description**
DC	Data center
PV	Photovoltaic
PUE	Power usage effectiveness
IT	Information technology
COP	Coefficient of performance

Functions

Symbol	**Description**
$L_{solar}(x)$	Solar suitability indicator at cell $x(0/1)$
$L_{wind}(x)$	Wind suitability indicator at cell $x(0/1)$
$\Psi(\mathrm{u})$	Wind turbine power curve vs. wind speed u
$U(d,h)$	Weekly load template (day d, hour h)
$M(m), \widehat{M}(m)$	Monthly load amplitude and normalized amplitude
$u(t)$	Normalized annual load shape
quantile $(\bullet,\alpha)$	α-quantile of a time series
clip(x,a,b)	Clipping to $[a,b]$
$h(T,w)$	Moist-air enthalpy at temperature
T, humidity ratio w	

Sets and indices

Symbol	**Description**
G	Set of candidate grid cells (DC/wind/PV)
$g \in G$	Index of wind/PV siting grid
$h \in G$	Index of DC / substation location
τ	Hour index
K	Set of transformer types
$k \in K$	Transformer type index
$N(t)$	Set of active servers at hour t
d	Day-of-week index $(0,\cdots,6)$
r	Hour-of-day index $(0,\cdots,23)$
m	Month index $(1,\cdots,12)$

Parameter

Symbol	**Description**
Φ_τ^{wind}	Wind capacity factor at time τ
ε_τ^{pan}	Plane-of-array irradiance at time τ
κ_τ^{sh}	PV shading coefficient at τ
κ_τ^{T}	PV temperature coefficient at τ
λ_{loss}	Aggregate PV system loss factor
P_{max}	Target peak DC site load (MW)
α	Quantile level for scaling factor s
D_t	DC demand at hour t (MW)
D^{max}	Annual peak DC demand (MW)
$T_{db}(t)$	Dry-bulb temperature at hour $t(^\circ C)$
$T_{dew}(t)$	Dew-point temperature at hour $t(^\circ C)$
$T_{wb}(t)$	Wet-bulb temperature at hour $t(^\circ C)$
$p(t)$	Atmospheric pressure
T_{econ}	Economizer threshold $(^\circ C)$
β_0,β_1,T_{ref}	COP regression coefficients
COP_{min}, COP_{max}	COP lower/upper bounds
$C_{aux}(t)$	Auxiliary power fraction at hour t
C_{pd}, C_{pv}	Specific heats of dry air and water vapor
$L_v(t)$	Latent heat of vaporization at T
$k^w_{pix,g}, k^s_{pix,g}$	Cost per wind/PV pixel at cell g
N^w_g, N^s_g	Max. wind/PV pixels at cell g
$\widehat{c}^{\,w}_g, \widehat{c}^{\,s}_g$	Capacity per wind/PV pixel (MW/pixel)
M^w_g, M^s_g	Big-M constants for wind/PV at g
$c^{bP}_{cap}, c^{bE}_{cap}$	Cost per battery capacity (annualised)
$\pi^{buy}_t, \pi^{sell}_t$	Buy/sell price at hour t
c_{spur}	Unit spur-line cost (wind/PV–DC)
c_{site}	Unit tie-line cost (DC–grid)
$d^{RE}_{h,g}$	Distance wind/PV cell g-DC cell h
d^{grid}_h	Distance DC cell h-nearest grid node
C_{site}	Fixed substation cost at DC site
C^{tr}_k	Cost per transformer of type k
C_{deg}	Battery degradation cost per throughput
η_{ch}, η_{dis}	Battery charge/discharge efficiency
Δt	Time step length (h)
F_{max}	Upper bound on grid exchange capacity (MW)
S_k	Apparent power of transformer type k (MVA)
M_k	Max. number of transformers of type k
PF_{min}	Minimum transformer power factor

Variables

Symbol	**Description**
G^{wind}_τ	Wind generation at time τ
K^{wind}	Installed wind capacity potential
G^{pv}_τ	PV generation at time τ
K^{pv}	Installed PV capacity potential
$p_{shape}(t)$	Raw measured DC power at hour t
$\widetilde{p}(t)$	Normalized DC power (dimensionless)
s	Scaling factor for DC load shape
$p_{site}(t)$	DC site load at hour t (MW)
$PUE(t)$	PUE at hour t
$P_{IT}(t)$	IT load at hour t (MW)
$P_{aux}(t)$	Auxiliary load at hour t (MW)
$P_{chiller}(t)$	Chiller power at hour t (MW)
$m(t)$	Mechanical cooling share at hour t
$COP(t)$	COP of chiller at hour t
$e(t), w(t)$	Vapor pressure and humidity ratio
C	Annualised total system cost (objective)
y^{dc}_h	1 if DC is located at cell h
y^w_g, y^s_g	1 if wind/PV built at cell g
n^w_g, n^s_g	Number of wind/PV pixels at cell g
C^w_g, C^s_g	Wind/PV capacity at cell g (MW)
$z^w_{h,g}, z^s_{h,g}$	Spur-line capacity from g to DC at h
f_h	Tie-line capacity attached to DC at h
P_b	Battery power rating (MW)
E_b	Battery energy capacity (MWh)
p^{buy}_t, p^{sell}_t	Grid buy/sell power at hour t (MW)
$\overline{p}^w_t, \overline{p}^s_t$	Available wind/PV power at hour t
p^{RE}_t	Renewable power delivered to DC at hour t
s_t	Renewable curtailment at hour t
$p^{ch.R}_t, p^{ch.G}_t$	Charge power of R/G warehouse at hour t
$p^{dis.R}_t, p^{dis.G}_t$	Discharge power of R/G warehouse at hour t
SoC^R_t, SoC^G_t	State of charge of R/G warehouse at hour t
z^{bat}_t	Battery charge/discharge mode (0/1)
F	Grid exchange capacity at DC bus (MW)
z^{grid}_t	Buy/sell mode at hour t (0/1)
δ_k	Transformer type selection (0/1)
n_k	Number of transformers of type k
S_{tot}	Total transformer capacity (MVA)
S_{unit}	Single-unit transformer capacity (MVA)
E^{avail}_{RE}	Annual available renewable energy (MWh)
E^{sell}_{RE}	Annual renewable energy exported (MWh)
E^{self}_{RE}	Annual renewable self-consumption (MWh)
E_{load}	Annual DC energy demand (MWh)

1. Introduction

1.1. Background and significance

The global energy transition is entering a phase of deep decarbonisation centred on power systems. Electricity is both the largest lever for achieving carbon neutrality [1] and the key enabling sector for decarbonising end uses in industry, transport, and buildings through electrification [2]. The power sector accounts for more than 40% of global CO_2 emissions from fuel combustion [3]. Meanwhile, demand has not declined organically. Electrification is increasing end-use reliance on grids, and the expansion of digital infrastructure is adding further upward pressure, together accelerating global electricity consumption in recent years. Global electricity demand was about 29,471 TWh in [4] (Ember [5]), and grew by roughly 4.3% year on year in [5] [6], exceeding the ~2.5% growth rate in [4] (Ember [5]). Data centers (DCs), as a new load with high utilisation and continuous operation, consumed around 415 TWh (about 1.5% of global electricity) and may rise to ~945 TWh by 2030, strengthening the structural drivers of demand growth [7]. Consequently, under expanding demand, simultaneously meeting requirements for reliable supply[8] and deep emissions reductions has become one of the most challenging tasks in power-system planning and operation [9].

Among the various drivers, the expansion of DCs is most clearly characterised by large-scale additions to electricity demand[10]. Driven by cloud computing and AI, DCs are evolving from supporting facilities of the information sector into a system-relevant, energy-intensive sector, featuring high load density, near-continuous operation, and strong spatial concentration[11]. Fig. 1 summarises the global concentration of DC deployment across the top 20 countries in terms of (a) facility counts, (b) estimated power capacity and renewable electricity share, and (c) hyperscale presence and growth rates [12,13]. This high-reliability electricity demand further amplifies the tension between reliable supply and deep decarbonisation.

Rapid renewable deployment is reshaping the supply side [14]. Over the past decade, sustained growth in wind and solar PV capacity, together with falling costs, has made these technologies economically competitive in an increasing number of regions and is shifting power systems from incremental supplementation to structural substitution [15]. However, as variable renewables reach higher shares, system constraints increasingly manifest as two forms of mismatch. A temporal mismatch means that “sufficient annual energy” can coexist with “insufficient hourly available power,” underscoring the need for storage and flexibility. Spatial and grid mismatches, in turn, separate resources from loads while exposing network bottlenecks, increasing congestion and curtailing risks [16]. Consequently, meeting annual energy quotas or relying on financial instruments such as renewable certificates does not automatically translate into verifiable system-wide emissions reductions. To deliver accountable abatement, planning must explicitly incorporate both hourly balancing and grid deliverability constraints, shifting from “annual compliance” to “hourly feasible” system solutions [17].

Under high shares of wind and solar PV, the scarce resource is often not renewable energy per se, but the ability to absorb it reliably on an hourly basis and to deliver it under network constraints. Relative to relying solely on long-distance transfers through the public grid, a more direct engineering approach is to steer electricity-sensitive and scalable loads toward resource-rich areas, enabling a spatial reconfiguration in which demand follows supply. China's “East Data, West Computing” initiative reflects this logic by reshaping the regional deployment of computing capacity and DCs, converting part of the cross-region power transfer pressure into cross-region flows of data and computing services [18]. Against this backdrop, a private wire supply architecture provides an actionable pathway to tightly couple renewables with high-reliability loads. As illustrated in Fig. 2, it can be viewed as an integrated energy hub that co-optimises supply, demand, and flexibility: wind and solar PV serve the load directly via dedicated lines; a battery energy storage system supports short term balancing and fluctuation damping; and a grid interconnection at the point of common coupling (PCC) preserves necessary energy exchange and backup support. Its core value lies in internalising integration and balancing on the load side, improving the usable availability of renewable output under PCC limits or congestion, and moving clean electricity procurement from annual matching toward closer hourly physical matching with accountable emissions reductions [9,18].

However, the effectiveness of a private wire arrangement is not guaranteed. It hinges on coordinated planning trade-offs: the scale and mix of wind and solar PV, the power and energy capacity of storage, and the exchange limits at the PCC must be determined jointly under reliability requirements and renewable electricity targets, with the electricity purchase and sale strategy also determined jointly. To address this, we develop a coordinated planning framework for private wire supplied DCs and formulate a mixed integer linear programming (MILP) model that jointly optimises wind, solar PV, storage, and grid interactions within a unified formulation, yielding interpretable, reproducible, and engineering-ready capacity configurations [19]. Policy developments also support this model. We compiled available national and selected provincial policy [20] documents in China that promote direct green power connections and specify requirements on source–load matching and interconnection management; a concise summary is provided in SI Table S1.

1.2. Related works

The energy challenge of DCs first manifests as a structural shift in load characteristics. Unlike conventional distributed commercial demand, DCs exhibit near-continuous operation, high load density, rapid growth, and strong spatial clustering, increasingly acting as power-demand units with system-level impacts[21]. In terms of end-use composition, total electricity consumption in DCs is typically divided into IT equipment (servers, storage, networking) and supporting infrastructure (cooling, power chain losses such as UPS and distribution, lighting, and auxiliary systems). The industry commonly uses power usage effectiveness (PUE) as the key indicator of infrastructure efficiency, defined as the ratio of total DC energy use to IT energy use. Within infrastructure loads, cooling often dominates both marginal efficiency and operational flexibility, and its performance depends on outdoor meteorological conditions, heat-rejection design, rack power density, and the chosen cooling technology. Existing reviews and engineering studies consistently show that cooling can account for a substantial share of total DC electricity use; thus, cooling strategies affect not only overall consumption but also reshape the load profile and the available flexibility [22].

Recent work has converged on a shared view regarding load flexibility: under certain conditions, DCs are not only energy-intensive loads but can also act as resources that provide demand response and system flexibility. This potential primarily comes through two pathways, the computing layer and the cooling layer. On the computing side, the core mechanism is workload shifting, which exploits task latency tolerance, task mobility, and queue scheduling constraints to rearrange computational demand over time or across locations while maintaining quality of service. Early demand response studies examined how workload shifting can reduce peaks, lower coincident peak risk, and cut operating costs, and explored coordination with on-site or backup generation [23]. As temporal and spatial variations in grid carbon intensity and renewable availability have been characterised more precisely, research has shifted toward carbon-aware scheduling, in which tasks are proactively shifted to lower-carbon periods or regions to reduce indirect emissions and better align with clean electricity. Google's Carbon Intelligent Computing illustrates an engineering implementable paradigm, using delay-tolerant workloads and intra-day carbon budgets to enable carbon-aware computing management [24]. More recent studies discuss

the contributions and limits of intertemporal and interregional shifting for 24/7 clean electricity matching under stricter hourly objectives [25].

Cooling-side flexibility mainly arises from the thermal inertia of buildings and cooling systems, as well as from decoupling enabled by thermal energy storage, such as cooling storage. Subject to temperature safety constraints, part of the cooling demand can be shifted away from high price or grid-stressed hours to more favourable periods, or used for short-term power shaping. In practice, data-driven control has already been used to reduce cooling energy use, with Google and DeepMind demonstrating intelligent control for energy savings, highlighting cooling as both a major energy use and a key lever for operational Optimisation [26]. In studies that emphasise dispatchability, chilled water storage, ice storage, air conditioning systems integrated with thermal storage, and even larger-scale seasonal cooling storage have been investigated to enhance peak shaving and load shifting, often coupled with grid demand response or price mechanisms [27,28]. Overall, while the literature has established a solid basis on end-use decomposition and flexibility mechanisms, whether these flexibility potentials can be converted into verifiable emissions reductions under stringent clean electricity goals still depends on the coordinated design of clean power supply, interconnection constraints, and system balancing resources.

For clean electricity procurement for DCs, the literature and practice broadly converge on two pathways. The first is a market-based, transaction-oriented approach, including corporate power purchase agreements (PPAs) and electricity environmental attribute certificates (EACs, such as RECs and GOs), which primarily support renewable electricity claims and Scope 2 accounting. The second is a physically grounded approach that prioritises engineering controllability, aiming to build stronger resource control and hourly matching capability on the load side. These two pathways respectively emphasise scalability and physical verifiability, yet both reveal structural limitations under high ambition targets [29]. Transaction-based procurement enables cross-entity and cross-project sourcing and hedging through accounting attribute assignment, and corporate PPAs can support financing and capacity build-out, but their mitigation effectiveness depends critically on additionality and the choice of geographic and temporal boundaries [30]. More fundamentally, Scope 2 "market-based" accounting can introduce systematic information bias. A well-known critique is that it can blur decision relevance and informational integrity, making "buying attributes" easily misinterpreted as "physical emissions from electricity use have been eliminated" [31]. Empirical evidence further suggests that, once certificate-driven "claimed reductions" are removed, the alignment of corporate Scope 2 trajectories with temperature goals may drop markedly, implying that certificate mechanisms can inflate the apparent performance of corporate decarbonisation [31]. Accordingly, the focus has been shifting from annual energy matching toward stricter constraints that explicitly consider time and grid deliverability. Recent studies decompose deliverability, additionality, and double counting, showing that even within the same grid and the same year, certificate accounting can decouple from physical consumption-based emissions, which calls for explicit treatment of deliverability and additionality in both accounting frameworks and procurement design [32]. Consistent with this direction, 24/7 carbon-free energy (24/7 CFE) and hourly matching are emerging as frontier approaches for continuous loads such as DCs. Google has proposed assessment frameworks that move beyond annual 100% claims to hourly matching, emphasising that annual offsetting does not imply carbon-free supply in every hour. Subsequent work quantifies pathways, cost premiums, and system impacts of 24/7 matching, indicating that hourly matching can more directly steer deployment toward critical hours and technologies under certain conditions[33], but it also substantially increases requirements for storage, dispatchable clean supply, and explicit handling of regional constraints [34]. Overall, transaction-based approaches remain advantageous in scalability and financial flexibility, yet their abstraction from time matching and network constraints creates a risk of divergence between "claims" and "physical delivery" at high target levels.

In parallel with transaction-based procurement, physically grounded supply models emphasise building controllable supply capability at the load site or in its near field, such as campus microgrids, co-located wind and solar PV with battery storage, and more engineered load-side energy hubs. Compared with certificate-based offsetting, these approaches more readily achieve verifiable physical matching, but their scalability is often constrained by siting availability, interconnection conditions, and the configuration of balancing resources. Existing studies have developed robust planning models for DC microgrids, or capacity planning and stochastic optimisation frameworks that jointly address thermal management and renewable integration, consistently showing that, when renewable variability and reliability requirements coexist, storage sizing and grid interaction strategies are decisive for both cost and feasibility [35]. Broader work on load side resources has also synthesized mechanisms and optimisation methods for behind-the-meter storage in peak shaving, resilience, and renewable integration, providing a transferable foundation for understanding endogenous balancing on the demand side [36]. This points to the need for an intermediate form that reconciles scalability with physical deliverability. At the modelling level, such a framework should integrate renewable variability, the dual dimensions of storage power and energy, PCC constraints, and procurement rules within a unified formulation, offering an engineering pathway for continuous loads such as DCs to move from "claims-based compliance" toward hourly accountable emissions reductions.

Private wire is emerging as a salient "third form" within this landscape. It differs from transaction-based procurement, such as PPAs and EACs, and also from on-site microgrids whose generation and consumption are tightly bound by a campus footprint. Its defining feature is an explicit physical delivery relationship between renewable generation and a single load via dedicated lines, while retaining necessary interaction with the public grid at the PCC. Because generation can be deployed in the load's near field without being strictly confined to the site boundary, this model is closer in engineering scale to utility-grade wind, solar PV, and storage deployment, while improving traceability and deliverability through a physical direct connection. It therefore offers a pragmatic compromise between scalability and physical matching [37]. Institutionally, EU electricity market rules define direct lines, providing a legal interface for direct connections between producers and consumers [38]. In the UK, regulatory and industry materials have long discussed the boundaries and cost differentials of private wire arrangements in metering, licensing, and commercial settlement [39]. Ireland has recently issued a Private Wires policy statement, highlighting its role in unlocking private investment in renewables and storage [40]. In China, "direct green power connections" have been promoted at the national level, defined as a model in which new energy supplies a single user via a dedicated line with physical traceability of delivered electricity [20].

Integrated source–grid–load–storage planning and microgrid optimisation offer a mature foundation for modelling private wire systems. Many studies use mixed integer linear programming (MILP) and related extensions to co-optimise wind, solar PV, and storage capacities together with hourly dispatch[41], while representing grid purchases and sales and their boundary conditions, thereby trading off capital expenditure against operational feasibility [42,43]. In parallel, energy system research for DCs is advancing rapidly, increasingly incorporating the schedulability of batch computing loads, thermal management equipment, and renewable generation uncertainty into planning and operational optimisation, underscoring the strong coupling between load characteristics and supply configurations [43].

Despite this progress, planning models for private wire supplied DCs still face several key gaps. First, although grid transactions are often modelled, the architecture specific system boundaries are rarely made explicit, including PCC capacity limits, the allocation between direct supply and grid export, reverse power flow restrictions, and responsibility interfaces consistent with metering and settlement rules.

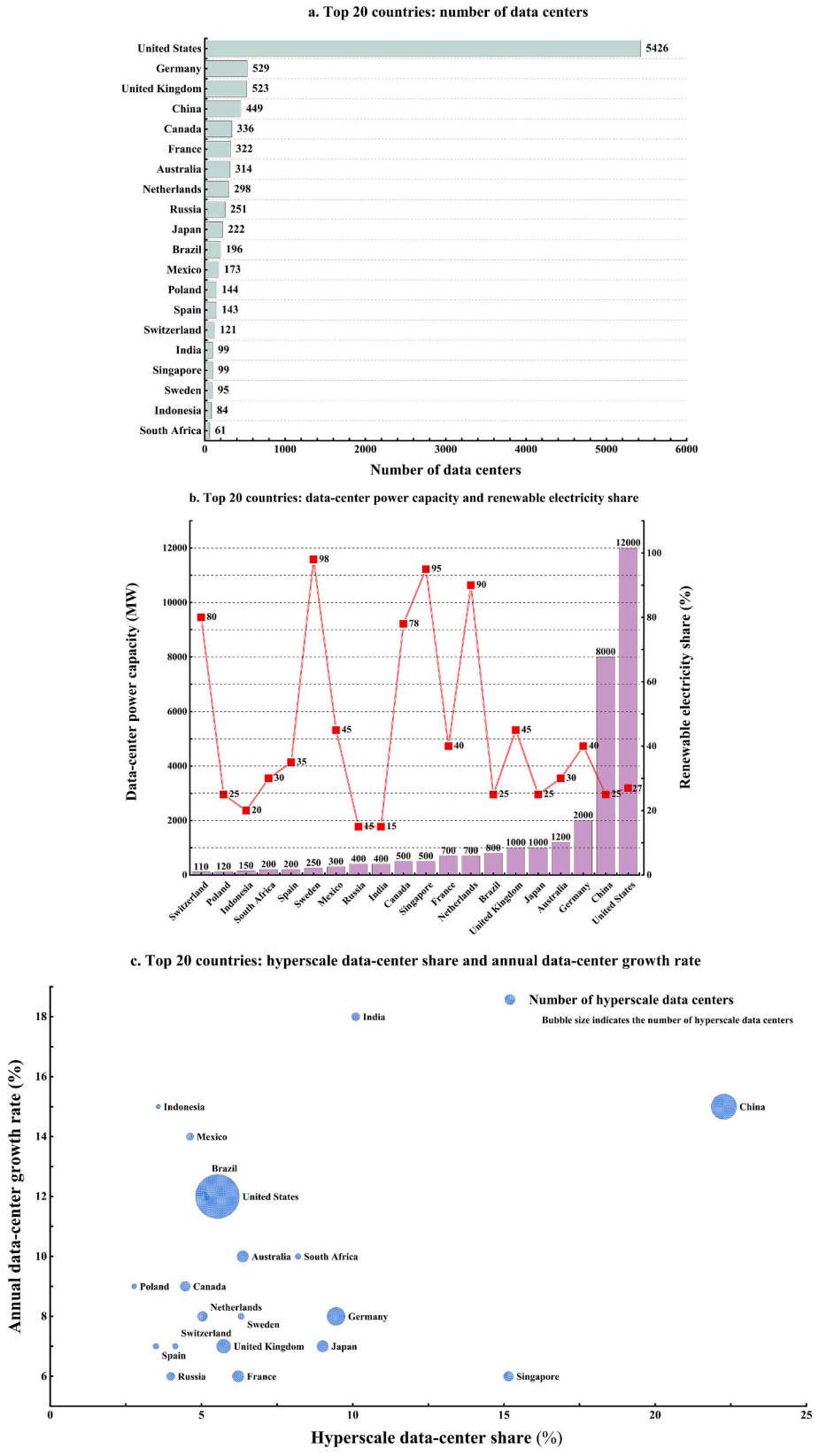


Fig. 1. Global concentration of DC deployment across the top 20 countries. (a) Number of DCs. (b) DC power capacity (MW) and renewable electricity share (%). (c) Hyperscale DC share (bubble size) and annual DC growth rate (%).

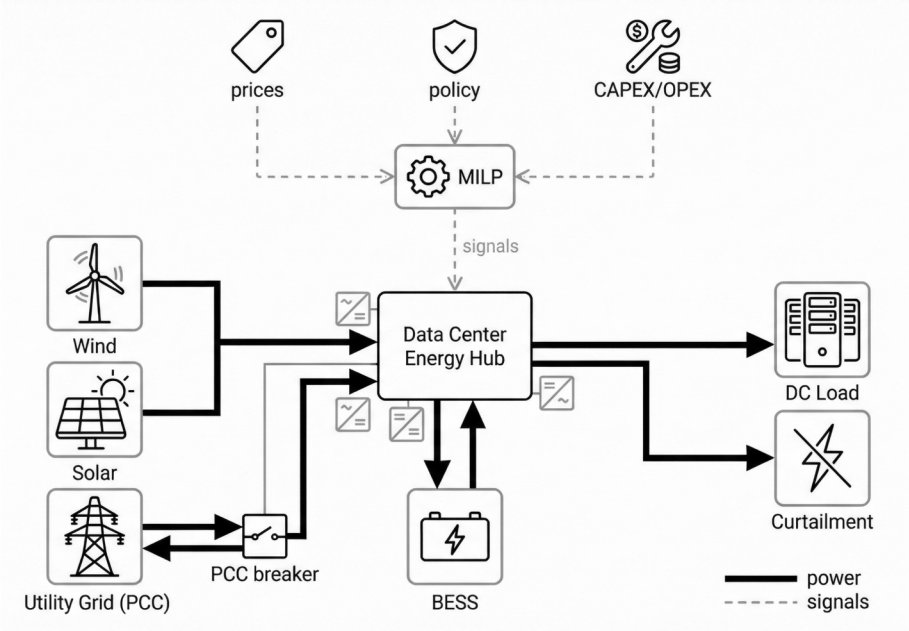


Fig. 2. Architecture of a private-wire renewable-supplied data-center energy hub.

This weakens the representation of congestion and deliverability under high renewable targets (NDRC and NEA, 2025). Second, the high reliability requirements of DCs are frequently treated via exogenous backup assumptions or simplified penalty terms, rather than through an integrated formulation that tightly links reliability to the renewable mix and the joint sizing of storage power and energy. Consequently, models struggle to answer a central question: what capacity premium is required to achieve hourly feasibility and reliability [35,43]. Third, the engineering value of direct supply typically relies on storage as a key mediator, yet storage power and energy sizing, charge and discharge strategies, and grid interactions are inherently coupled decisions that must be solved jointly with renewables and PCC constraints at the planning stage [42]. Therefore, the next step for private wire research is to move beyond conceptual discussions and point economic assessments toward a coordinated planning framework that simultaneously captures renewable variability, two-dimensional storage sizing, PCC constraints, and hourly energy balance, while endogenising high reliability load requirements within the model.

In summary, prior work on clean electricity supply for DCs has converged along three strands (Table 1). First, studies on end-use decomposition and flexibility show that workload shifting and cooling-side flexibility can provide demand response capability. Second, transaction-based procurement is widely used, but annual attribute matching can decouple from hour-by-hour physical delivery, potentially leading to a divergence between "claims" and "physical delivery" at high ambition levels. Third, near-field microgrids and on-site self-supply improve physical controllability, yet are often constrained by land availability, interconnection conditions, and balancing requirements for hyperscale loads. These strands collectively motivate a central gap: private-wire arrangements can reconcile scalability with physical delivery, but DC-oriented planning models still lack a unified representation of renewable variability, two-dimensional battery sizing (power and energy), PCC constraints, and electricity trading rules under stringent reliability requirements. Building on this motivation, this study addresses the following research questions:

(1) Under a private wire architecture with PCC constraints, how can a high share of clean electricity supply be achieved with hourly physical deliverability, while avoiding a decoupling between claims-based compliance and real emissions reductions?
(2) How should wind, solar PV, and battery storage capacities be co-determined, and how does the optimal combination of storage power and energy change with clean electricity targets and interconnection conditions?
(3) Once high reliability requirements and hourly energy balance are enforced for DCs, how do the dominant cost drivers shift, and what roles are played by grid interactions versus storage-based balancing?
(4) Across scenarios, what emissions reductions and marginal costs are achievable, and what are the binding bottlenecks: renewable variability, grid deliverability, or storage configuration?

1.1. Novelty and contribution

This paper develops an integrated framework spanning physical modelling, coordinated planning, and value assessment. The central innovation is to use a private wire architecture to move clean electricity supply for DCs from annual attribute claims toward hourly, physically deliverable provision, while endogenising renewable variability, the two-dimensional sizing of battery storage (power and energy), and PCC interaction constraints within a unified planning model. Our contributions are threefold:

(1) City-scale source-load co-siting for new-build private-wire DCs. We formulate a spatially explicit planning problem that jointly optimises the siting of a new DC hub and the matched wind, PV, storage, and interconnection assets within the city-level decision boundary, enabling an actionable recommendation on where a future private-wire DC should be built and which local supply grids should be developed.

(2) Policy-grounded pathways via multi-dimensional scenario analysis. Building on the current policy baseline and progressive tightening cases, we provide a consistent planning envelope that characterizes how optimal capacity mixes, grid interaction roles, and cost structures evolve as clean-supply targets tighten, offering decision-relevant pathways for both present deployment and higher-share transition planning under direct-connection rules.

(3) A two-dimensional storage planning and evaluation contribution. By co-optimising storage power and energy and tracking feasibility under 8760-h deliverability constraints, we show why storage should be assessed in both dimensions, and why energy-only metrics or annual throughput can be misleading: If evaluation focuses only on storage energy or annual throughput, it may underestimate hourly feasibility

constraints, thereby amplifying cost jumps and feasibility risks as targets tighten.

The remainder of the paper is organised as follows. Section 2 describes the problem formulation, model, and data. Section 3 presents the case study results and analyses differences in capacity configuration and economics. Section 4 discusses engineering and policy implications. Section 5 concludes and outlines directions for future work.

2. Data and methods

2.1. Study area

This study uses Qinghai Province (Fig. 3) as an illustrative case region to demonstrate the proposed planning framework, as it jointly satisfies several conditions central to private wire-supplied data centres.

First, Qinghai spans nationally recognised high-quality wind and solar resource corridors, particularly along the Qaidam Basin and the Haixi corridor. ERA5-based climatological layers (2005–2024) indicate that the mean annual total GHI ranges from 1723 to 2286 kWh m^{-2} yr^{-1} across the province, substantially above the China-wide mean annual GHI reported in national resource assessments (~1491 kWh m^{-2} yr^{-1} [44]). Consistent with national atlas products, Qinghai is widely identified as one of China's most abundant solar-resource provinces [45], and the Qaidam Basin is classified among the highest-abundance zones within the province. For wind, the multi-year mean 100 m wind speed reaches up to 6.51 m s^{-1} in the strongest corridors, exceeding the 6 m s^{-1} (100 m) level often used as a screening rule-of-thumb in utility-scale wind siting assessments [46]. More broadly, the literature characterizes northwestern China and the northeastern Tibetan Plateau as a high-quality wind belt [47].

Second, Qinghai's high elevation and arid climate imply persistently low wet-bulb temperatures and high availability of economiser (free-cooling) operation [48]. Prior data-centre cooling studies show that free-cooling feasibility is tightly constrained by wet-bulb temperature, and best-practice guidance notes that cool climates can enable extensive economiser-based cooling, thereby reducing mechanical cooling demand and improving PUE.

Third, Qinghai's transmission system exhibits a corridor-style structure that is well-suited for explicitly representing deliverability limits and export costs in an MILP. Notably, the Qinghai–Tibet interconnection is built around 750 kV AC corridors linking Xining and Golmud, and recent 750 kV projects in resource-rich prefectures have been commissioned to integrate large wind and PV bases into the grid [49]. We emphasise that the case study is used to demonstrate implementability with realistic geography, meteorology, and network structure; the formulation and workflow are transferable to other resource-rich regions.

2.2. Data

The dataset comprises four categories: land and environmental constraints, meteorological reanalysis, infrastructure data, and measured DC data. For land and environmental constraints, we use a land cover dataset [50] to identify developable land, and exclude ecological redlines and sensitive areas by integrating WDPA layers [51], ecological function zoning [52], and World Heritage site datasets [53]. Geological and soil background data [54,55] are used to construct constraints or sensitivity analyses related to seismic risk and foundation conditions. Meteorological inputs are taken from ERA5 reanalysis [56] to simulate hourly wind and solar PV generation and to compute PUE; the key variables are summarised in Table 2. Key fixed technology and policy parameters used in the wind-PV-storage modelling and the optimisation constraints are summarised in SI Table S3. Grid and infrastructure data [57] are derived from Open Infrastructure Map, including transmission lines and substations, and are used to parameterise interconnection distances and access costs. On the DC side, we use measured trace data provided by UniMelb [58] to characterise load locations and operating patterns, and to support the specification of load constraints and plausibility checks for results under direct supply scenarios. We adopt an hourly (8760 h) temporal resolution because the study is planning-oriented and focuses on investment siting and sizing under meteorology-driven constraints on renewable availability and deliverability. Sub-hourly DC load fluctuations and fast operational dynamics are important for real-time control, but they are beyond the scope of this capacity-planning model and are not expected to alter the main planning-level conclusions regarding siting patterns, capacity mix, and the shift from grid-trading-dominated balancing to capital-dominated near self-sufficiency. Extensions to finer time steps are feasible as future work, but would require higher-resolution meteorological and load inputs and substantially increase computational burden.

2.3. Methods

Fig. 4 summarises the workflow of this study. First, we develop a wind and solar PV siting suitability assessment and screen developable land by integrating topography, land use, and ecological constraints; in parallel, prefecture level cities and autonomous prefectures are aligned to a 0.25° regular grid consistent with ERA5. Second, within suitable units, we specify representative turbine and PV configurations and deployment densities, and simulate hourly electricity generation over a full year for wind turbines and PV arrays using meteorology-driven models[59,60]. These siting and generation steps follow established, widely used engineering practice; to keep the main text concise, the detailed formulations and parameter settings are provided in the SI (Supporting Methods S1.1–S1.2). Third, we construct annual hourly electricity demand profiles for a standardised new-build hyperscale DC using historical load shapes and a ‘shape–capacity decoupling’ approach, with climate-driven PUE translating IT load into total facility demand. Finally, we formulate a load-following coordinated optimisation model for wind–solar–storage capacity and operations. Given the fixed DC demand, reliability constraints, and private-wire deliverability limits, the model determines the optimal host location (private-wire hub) and the matched wind/PV capacities, storage sizing, and operating strategies.

2.3.1. Data center load model

Because hyperscale DCs are relatively few and their operational data are constrained by confidentiality and compliance requirements, publicly available long-term, high-resolution, end-use disaggregated measurements that jointly cover “IT load, cooling, and total facility power” remain extremely scarce. We therefore adopt a reconstruction strategy that combines shape and capacity decoupling with a meteorology-coupled PUE model. Specifically, we extract a representative service demand shape from measured traces of the UniMelb DC [58], scale it to a target IT capacity using an engineering rack-based benchmark, and infer hourly PUE from ERA5 2 m dry bulb and dew point temperatures. This yields annual (8760 h) time series for IT power, cooling power, and total facility power.

First, we align server-level power to hourly timestamps, sum across active servers, and convert to MW to obtain the site-level hourly power shape:

$$p_{shape}(t) = \frac{\sum_{n \in \mathcal{N}(t)} \overline{P_n}(t)}{10^6} \ [MW] \tag{1}$$

Where $p_{shape}(t)$ is site power at hour t, $\mathcal{N}(t)$ is the set of active servers during hour t, and $\overline{P_n}(t)$ is the average power of server n in that hour (W). To reduce the influence of extremes, we normalise using the historical 95th percentile:

$$\widetilde{p}(t) = \frac{p_{shape}(t)}{q_{95}} \tag{2}$$

Table 1
Comparison of research strands on clean electricity supply for DCs and the positioning of this study.

Aspect	Strand 1: Flexibility and end-use decomposition	Strand 2: Transaction-based procurement and 24/7 matching	Strand 3: Near-field microgrids and on-site self-supply	This study
Main aim	demand flexibility for cost or emissions	clean-electricity procurement and hourly matching	physical self-supply near the load	hour-by-hour deliverable clean supply under project boundaries
System boundary	DC operations; grid exogenous	accounting boundary; physical delivery abstracted	campus or near-field; simplified grid interface	dedicated-line deliverability and PCC exchange limit
Key decisions	workload shifting; cooling control; TES	PPAs/EACs; matching rules; boundary choices	PV/wind siting; storage sizing; dispatch	DC siting; wind/PV build-out; battery power and energy; lines; trading; N − 1
Typical limitations	no PCC cap; no direct-line deliverability; no export/self-use rules	no deliverability and congestion constraints; "claims to delivery" risk	private-wire architecture not explicit; hyperscale boundary limits	unified 8760-h feasibility with PCC, deliverability, trading rules, and N − 1
Representative studies	([23]; [27]; [24]; [25])	([31]; [29]; [33]; [34])	([41]; [35]; [36]; [42])	–

$$q_{95} = \text{quantile}\left(p_{shape}(t), 0.95\right) \tag{3}$$

Where $\tilde{p}(t)$ is a dimensionless relative intensity and q_{95} is the 95th percentile site power (MW).

We then extract a weekly rhythm and a seasonal amplitude from $\tilde{p}(t)$, and combine them to form an annualised 8760 h profile:

$$U(d,h) = median\{\tilde{p}(t) \mid dow(t) = d, hour(t) = h\} \tag{4}$$

$$M(m) = median\{\tilde{p}(t) \mid \text{month(t)} = \text{m}\}, \widehat{M}(m) = \frac{M(m)}{median_m M(m)} \tag{5}$$

$$u(t) = \frac{U(dow(t), hour(t)) \bullet \widehat{M}(month(t))}{p_{99}}, p_{99} = \text{quantile}(U \bullet \widehat{M}, 0.99) \tag{6}$$

Here, $U(d,h)$ is a weekly template indexed by day of week $d \in \{0, \ldots, 6\}$ and hour $h \in \{0, \ldots, 23\}$; $M(m)$ and $\widehat{M}(m)$ capture the monthly median amplitude and its normalisation; $u(t) \in [0,1]$ is the annualised, dimensionless load shape, and p_{99} suppresses residual spikes.

We scale the dimensionless shape to an engineering benchmark peak site load p_{max} for a hyperscale facility (25,000 racks) via a single parameter:

$$s = \frac{p_{max}}{\text{quantile}(u(t), \alpha)} \tag{7}$$

$$p_{site}(t) = s \bullet u(t)\ [MW] \tag{8}$$

Where s is a dimensionless scaling factor, α is a chosen upper quantile for peak alignment, and $p_{site}(t)$ is the hourly site level load (MW).

To reconstruct hourly PUE, we back calculate PUE using climate driven cooling performance [61,62]:

$$PUE(t) = 1 + \frac{m(t)}{COP(t)} + c_{aux}(t) \tag{9}$$

$$P_{IT}(t) = \frac{P_{site}(t)}{PUE(t)}, P_{aux}(t) = c_{aux}(t) \times P_{IT}(t), P_{chiller}(t) = \frac{m(t) \times P_{IT}(t)}{COP(t)} \tag{10}$$

Here, $m(t) \in [0,1]$ is the fraction of mechanical cooling (0 for full free cooling, 1 for fully mechanical cooling), driven by wet bulb temperature; $COP(t)$ is the chiller coefficient of performance (cooling output over electric input), temperature dependent; and $c_{aux}(t)$ represents parasitic power (fans, pumps, cooling towers) as a fraction of IT power.

Following (Zhang et al. [5]), the mechanical cooling share is:

$$m(t) = clip\left(\frac{T_{wb}(t) - T_{econ}}{\Delta T}, 0, 1\right) \tag{11}$$

Where $T_{wb}(t)$ is ambient wet bulb temperature and T_{econ} is the economiser threshold. Chiller performance is approximated by [63]:

$$COP(t) = clip\left(\beta_0 + \beta_1\left[T_{ref} - T_{wb}(t)\right], COP_{min}, COP_{max}\right) \tag{12}$$

Where β_0,β_1,T_{ref} are system specific empirical parameters. The parasitic term $c_{aux}(t)$ is typically treated as a constant in [0.02,0.08].

Wet bulb temperature $T_{wb}(t)$ is computed from dry bulb temperature $T_{db}(t)$, dew point $T_{dew}(t)$, and surface pressure $p(t)$ using psychrometric relations [64]. Saturation vapor pressure over water and ice is:

$$e_s^{(w)}(T) = 6.112\exp\left(\frac{17.62T}{243.12+T}\right), e_s^{(i)}(T) = 6.112\exp\left(\frac{22.46T}{272.62+T}\right) \tag{13}$$

$$e(t) = 100 \times \begin{cases} e_s^{(w)}(T_{dew}(t)), & T_{dew} \geq 0^{\circ}C \\ e_s^{(i)}(T_{dew}(t)), & T_{dew} < 0^{\circ}C \end{cases} \tag{14}$$

Humidity ratio and saturated humidity ratio are:

$$w(t) = \frac{0.62198e(t)}{p(t) - e(t)} \tag{15}$$

$$w_s(T;p) = \frac{0.62198e_s(t)}{p - e_s(t)}, e_s(T) = 100 \times \begin{cases} e_s^{(w)}(T), & T \geq 0^{\circ}C \\ e_s^{(i)}(T), & T < 0^{\circ}C \end{cases} \tag{16}$$

Moist air enthalpy is:

$$h(T,w) = c_{pd}T + w\left(L_v(T) + c_{pv}T\right) \tag{17}$$

With
$c_{pd} = 1.006, c_{pv} = 1.86$, (and)

$$hL_v(T) = 2500.9 - 2.381T \tag{18}$$

Finally, $T_{wb}(t)$ is obtained by solving the isobaric adiabatic humidification enthalpy balance [65]:

$$h(T_{db}(t), w(t)) = h(T_{wb}(t), w_s((T_{wb}(t); p(t)))) \tag{19}$$

2.3.2. *Wind–solar–storage–grid–compute co-optimisation model*

We formulate an MILP to co-optimise private-wire supplied wind-solar-storage-grid systems for DCs under hourly constraints. Decision variables determine (i) the siting and sizing of wind/PV and the private-wire interconnection and dedicated lines, (ii) the sizing and hourly operation of battery storage, and (iii) hourly grid purchase/sale decisions (see Nomenclature: Variables). Fixed inputs include the hourly DC IT-load profile and climate-driven PUE; meteorology-driven wind/PV availability profiles; land-buildability masks; distances/topology and unit costs; electricity price parameters; and policy targets. Co-located wind, solar PV, and battery storage developed as an integrated project for DCs is an increasingly important pathway. This study minimises the annualised total cost of building and operating the wind, solar PV, storage, and grid connection system serving a DC. The objective is to minimise the annualised total cost C_{total}:

$$\min C_{total} \tag{20}$$

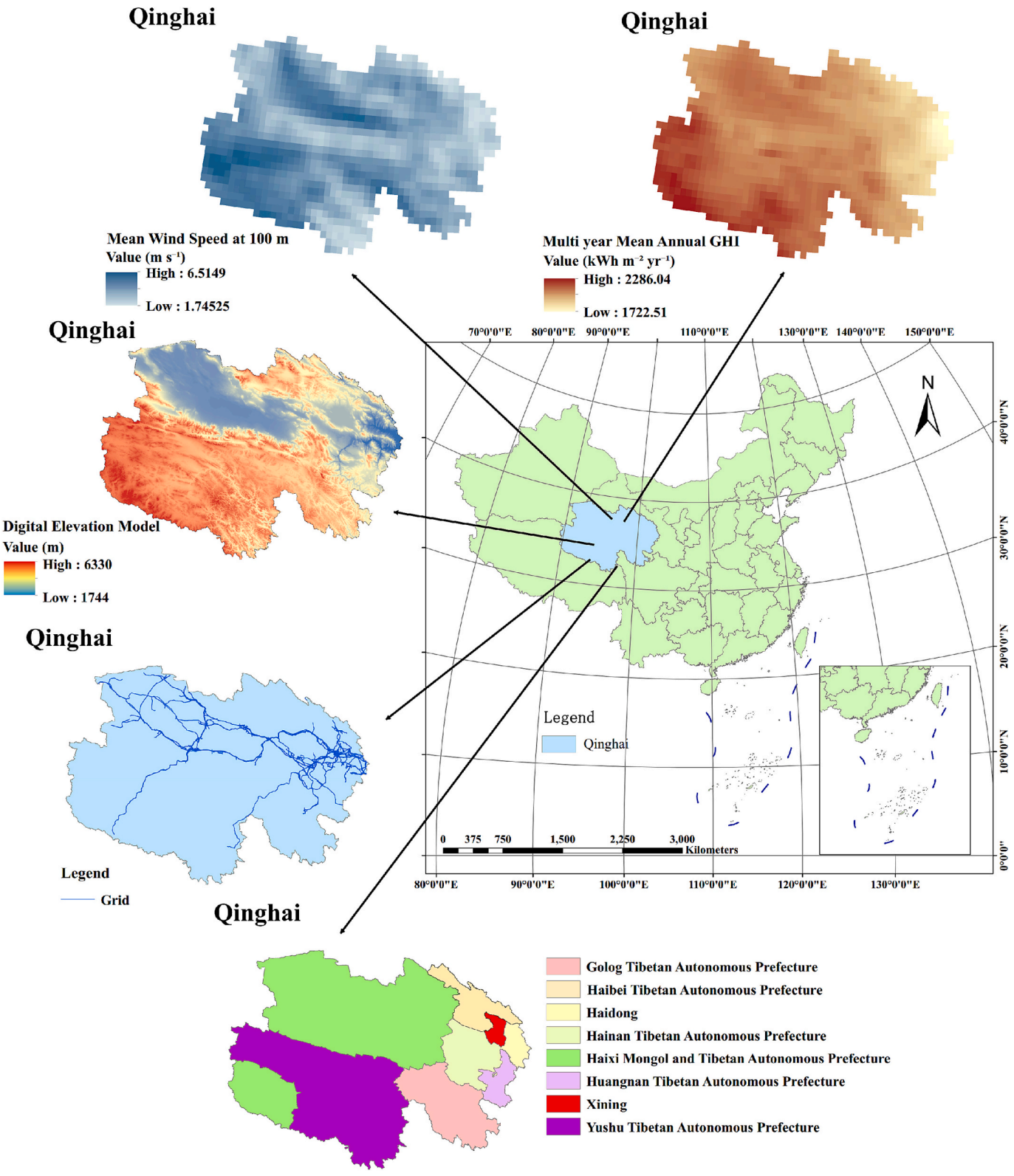


Fig. 3. Study area.

Table 2
ERA5 meteorological variables used for wind, PV, and PUE modelling.

Weather Factors	Description
U_{100}	100 m u-component of wind
V_{100}	100 m v-component of wind
S_{direct}	Total sky direct solar radiation at surface
S_{diff}	Surface solar radiation downward, clear sky
S_{total}	Surface net solar radiation
T_{atm}	2 m dry bulb air temperature
T_{dew}	2 m dry dew point temperature
P_{sfc}	Surface pressure

$$C_{total} = \underbrace{\sum_{g\in G}\left(k^{w}_{pix,g}n^{w}_{g} + k^{s}_{pix,g}n^{s}_{g}\right) + c^{bP}_{cap}P_b + c^{bE}_{cap}E_b}_{\text{Source and storage CAPEX}} + \underbrace{\sum_{t\in T}\left(\pi^{buy}_{t}p^{buy}_{t} - \pi^{sell}_{t}p^{sell}_{t}\right)}_{\text{Net grid–trading cost}}$$
$$+ \underbrace{c_{spur}\sum_{h\in G}\sum_{g\in G}d^{RE}_{h,g}\left(z^{w}_{h,g} + z^{s}_{h,g}\right) + c_{tie}\sum_{h\in G}d^{grid}_{h}f_h + C_{site}\sum\nolimits_{h} y^{dc}_{h} + \sum_{k\in K}C^{tr}_{k}n_k}_{\text{Network and substation CAPEX}}$$
$$+ \underbrace{C_{deg}\sum_{t\in T}\left(\frac{p^{ch,R}_{t} + p^{ch,G}_{t} + p^{dis,R}_{t} + p^{dis,G}_{t}}{2}\right)\Delta t}_{\text{Battery degradation cost}} \tag{21}$$

Here, $C_{total}$is the annualised total cost. It comprises four parts: (i) source and storage CAPEX, including wind/PV build-out and battery power and energy capacity costs; (ii) net grid-trading cost at the PCC; (iii) network and substation CAPEX, including dedicated renewable spur lines, the tie-line to the grid interconnection point, and substation/transformer assets; and (iv) battery degradation cost. Specifically, n^{w}_{g} and n^{s}_{g} denote the number of wind or PV pixels built in grid cell g, with discounted per pixel costs $k^{w}_{pix,g}$ and $k^{s}_{pix,g}$. P_b and E_b are the battery rated power and energy capacity, with discounted costs c^{bP}_{cap} and c^{bE}_{cap}. π^{buy}_{t}, p^{buy}_{t}, π^{sell}_{t}, $p^{sell}_{t}$are the purchase and sale prices and power exchanged with the grid at hour t. $d^{RE}_{h,g}$ and d^{grid}_{h} are distances from renewable sites g to the candidate DC site h, and from DC site h to the nearest point of interconnection, respectively. $z^{w}_{h,g}$, $z^{s}_{h,g}$, and f_h are the equivalent capacities of the dedicated lines from renewables to the DC and the interconnection capacity to the grid, with discounted unit costs c_{spur} and c_{tie}. C_{site} and C^{tr}_{k} are the fixed cost of the substation and the discounted cost of a transformer of type k, and n_k is the number of transformers of type k. To facilitate later constraints on renewable self-use and export, the battery is split into a renewables compartment R and a grid charged compartment G. $p^{ch,R}_{t}$, $p^{ch,G}_{t}$, $p^{dis,R}_{t}$, $p^{dis,G}_{t}$ are the charging and discharging power in the two compartments, and C_{deg} is an energy throughput-based degradation cost. Battery power and energy are co-optimised as continuous decision variables. In practical deployment, these capacities can be translated into discrete selections of battery containers and inverter modules, but such discretisation is not explicitly modelled because it does not alter the planning-level conclusions targeted here. Storage duration is not imposed as a discrete candidate class; instead, it is determined endogenously by the optimised energy-to-power ratio E_b/P_b. For presentation, reported duration values are rounded to the nearest integer hour. All continuous decision variables are nonnegative.

The remaining constraints are as follows. The constraints are presented in the same order as the system's physical and policy logic. Eqs. (22)–(24) define DC siting, technology exclusivity, and the mapping from developable pixels to installed wind/PV capacity. Eqs. (25)–(28) enforce hourly renewable availability, curtailment accounting, and nodal power balance under the private-wire boundary. Eqs. (29)–(33) capture battery state-of-charge dynamics, operating limits, and mutually exclusive charge/discharge. Eq. (34) limits grid exchange capacity and prevents simultaneous buying and selling. Eqs. (35)–(37) enforce substation sizing and N-1 security. Eqs. (38)–(40) linearise the coupling between siting, dedicated-line capacity, and PCC interconnection; finally, Eqs. (41)–(43) implement policy constraints for direct green

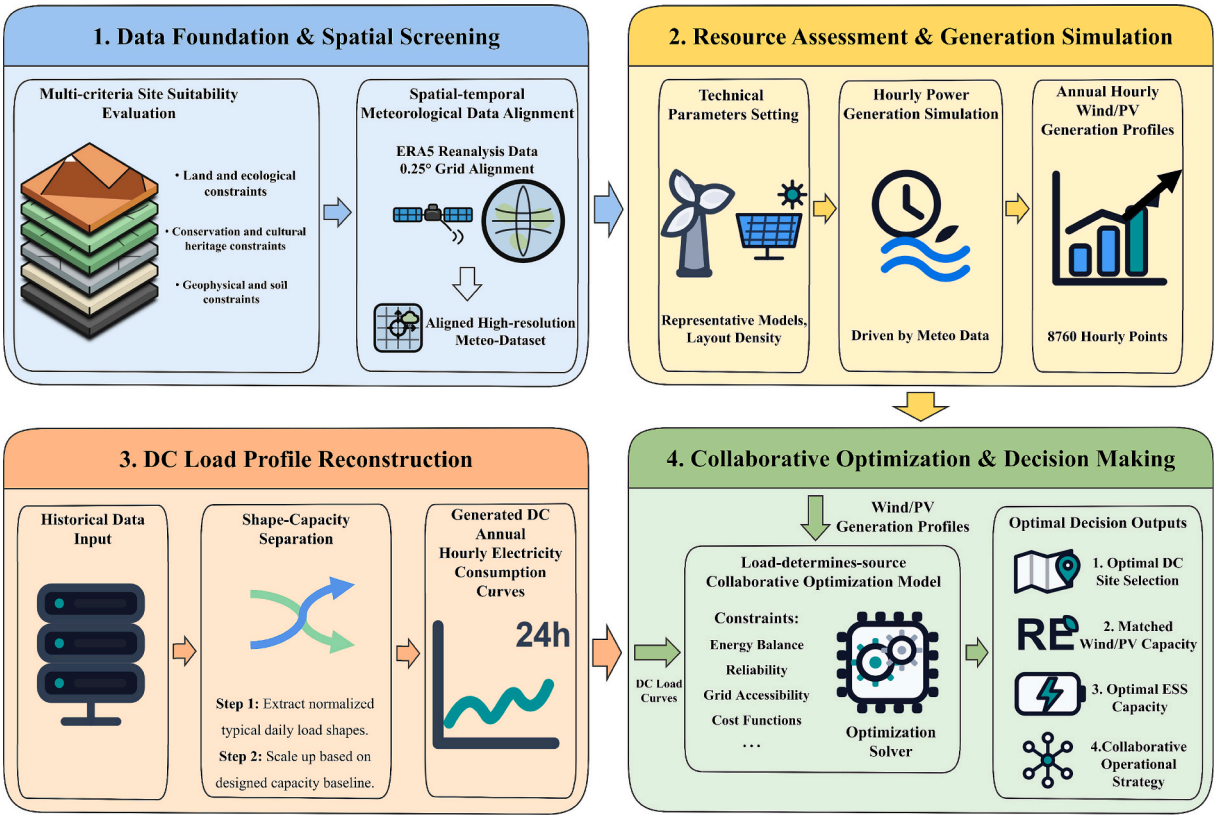


Fig. 4. Research framework.

power connections. A conventional reserve-margin constraint is not included because adequacy is enforced directly at the project boundary via the 8760-h balance under the PCC exchange limit, together with substation sizing and N-1 security.

Single **DC** siting, technology exclusivity, and pixel to capacity mapping

$$\sum_{h\epsilon G} y_h^{dc} = 1, y_h^{dc} \in \{0,1\}, \forall h \tag{22}$$

To avoid double occupation of land and corridor capacity within the same grid and to keep the model tractable, we assign each grid cell a dominant development type, wind or PV:

$$y_g^w + y_g^s \le 1, y_g^w, y_g^s \in \{0,1\}, \forall g \tag{23}$$

$$0 \le \mathrm{n_g^w} \le \mathrm{N_g^w y_g^w}, 0 \le \mathrm{n_g^s} \le \mathrm{N_g^s y_g^s}, \mathrm{n_g^w}, \mathrm{n_g^s} \in \mathbb{Z} \ge 0, \forall \mathrm{g},$$

$$C_g^w = \hat{c}_g^w n_g^w, C_g^s = \hat{c}_g^s n_g^s, \forall g. \tag{24}$$

Hourly renewable availability, curtailment, and dedicated line capacity

$$\overline{\mathrm{p}}_\mathrm{t}^\mathrm{w} = \sum_{\mathrm{g}\in\mathrm{G}} \mathrm{n_g^w} \hat{\mathrm{p}}_{\mathrm{g,t}}^\mathrm{w}, \overline{\mathrm{p}}_\mathrm{t}^\mathrm{s} = \sum_{\mathrm{g}\in\mathrm{G}} \mathrm{n_g^s} \hat{\mathrm{p}}_{\mathrm{g,t}}^\mathrm{s}, \forall \mathrm{t}$$

$$p_t^{RE} + s_t = \overline{p}_t^w + \overline{p}_t^s, p_t^{RE} \ge 0, s_t \ge 0, \forall t \tag{25}$$

$$p_t^{RE} \le \sum_{h\epsilon G} \sum_{g\epsilon G} \left(z_{h,g}^w + z_{h,g}^s \right), \forall t \tag{26}$$

Power balance and renewable flow restrictions

$$p_t^{RE} + p_t^{dis,R} + p_t^{dis,G} + p_t^{buy} = D_t + p_t^{ch,R} + p_t^{ch,G} + p_t^{sell}, \forall t \tag{27}$$

$$\mathrm{p_t^{ch,R}} \le \mathrm{p_t^{RE}}, \forall \mathrm{t}$$

$$\mathrm{p_t^{sell}} \le \mathrm{p_t^{RE}} + \mathrm{p_t^{dis,R}}, \forall \mathrm{t}$$

$$p_t^{ch,G} \le p_t^{buy}, \forall t \tag{28}$$

State of charge dynamics and limits

$$SoC_{t+1}^R = SoC_t^R + \eta_{ch} p_t^{ch,R} \Delta t - \frac{1}{\eta_{dis}} p_t^{dis,R} \Delta t,$$

$$SoC_{t+1}^G = SoC_t^G + \eta_{ch} p_t^{ch,G} \Delta t - \frac{1}{\eta_{dis}} p_t^{dis,G} \Delta t, \forall t \tag{29}$$

$$\mathrm{SoC_t^R} \ge 0, \mathrm{SoC_t^G} \ge 0, \mathrm{SoC_t^R} + \mathrm{SoC_t^G} \le \mathrm{E_b}, \forall \mathrm{t},$$

$$0 \le p_t^{ch,R} + p_t^{ch,G} \le P_b, 0 \le p_t^{dis,R} + p_t^{dis,G} \le P_b, \forall t. \tag{30}$$

Cycle consistency and no simultaneous charge and discharge:

$$SoC_1^R = SoC_{|T|}^R, SoC_1^G = SoC_{|T|}^G \tag{31}$$

$$p_t^{ch,R} + p_t^{ch,G} \le P_b z_t^{bat}, p_t^{dis,R} + p_t^{dis,G} \le P_b \left(1 - z_t^{bat}\right), z_t^{bat} \in \{0,1\}, \forall t. \tag{32}$$

Grid exchange capacity and mutual exclusivity of buying and selling

$$0 \le p_t^{buy} \le F, 0 \le p_t^{sell} \le F, 0 \le F \le F_{max}, \forall t, \tag{33}$$

$$p_t^{buy} \le F z_t^{grid}, p_t^{sell} \le F\left(1 - z_t^{grid}\right), z_t^{grid} \in \{0,1\}, \forall t. \tag{34}$$

Substation capacity and N-1 security constraints:

$$\sum_{k\in K} \delta_k = 1, 0 \le n_k \le M_k \delta_k, \delta_k \in \{0,1\}, n_k \in \mathbb{Z} \ge 0, \forall k, \tag{35}$$

$$PF_{min} S_{tot} \ge D^{max} + F \tag{36}$$

$$PF_{min}(S_{tot} - S_{unit}) \ge D^{max} + F, \sum_k n_k \ge 2 \tag{37}$$

$$0 \le F \le F_{max} \tag{38}$$

Following engineering practice for hyperscale DCs, we enforce substation sizing and N-1 security through Eqs. (35)–(37). Specifically, Eq. (35) enforces the selection of a single transformer type, ensuring that installed units are identical; Eq. (36) requires the total installed transformer capacity to satisfy the peak demand plus PCC exchange capacity; and Eq. (37) implements the N-1 criterion by requiring the remaining available transformer capacity after the loss of one unit to still cover the same requirement, while also enforcing at least two installed units.

Linearisation of dedicated line capacity and interconnection with **DC** siting.

Let $M_g^w = \hat{c}_g^w N_g^w, M_g^s = \hat{c}_g^s N_g^s$. For all $h, g \in G$:

$$0 \le \mathrm{z_{h,g}^w} \le \mathrm{C_g^w}, \mathrm{z_{h,g}^w} \le \mathrm{M_g^w y_h^{dc}}, \mathrm{z_{h,g}^w} \ge \mathrm{C_g^w} - \mathrm{M_g^w}\left(1 - \mathrm{y_h^{dc}}\right),$$

$$0 \le z_{h,g}^s \le C_g^s, z_{h,g}^s \le M_g^s y_h^{dc}, z_{h,g}^s \ge C_g^s - M_g^s \left(1 - y_h^{dc}\right). \tag{39}$$

Interconnection capacity coupled with **DC** siting:

$$0 \le f_h \le F, f_h \le F_{max} y_h^{dc}, f_h \ge F - F_{max}\left(1 - y_h^{dc}\right), \forall h. \tag{40}$$

Policy constraints for direct green power connections

$$E_{RE}^{avail} = \sum_{t\in T} \left(\overline{p}_t^w + \overline{p}_t^s\right) \Delta t, E_{RE}^{sell} = \sum_{t\in T} p_t^{sell} \Delta t, \tag{41}$$

$$E_{RE}^{self} = \sum_{t\in T} \left[p_t^{RE} - p_t^{sell} + p_t^{dis,R}\right] \Delta t, E_{load} = \sum_{t\in T} D_t \Delta t, \tag{42}$$

$$E_{RE}^{self} \ge 0.6 E_{RE}^{avail}, E_{RE}^{self} \ge 0.3 E_{load}, E_{RE}^{sell} \le 0.2 E_{RE}^{avail}. \tag{43}$$

Renewable variability is represented explicitly by the 8760-h wind/PV availability profiles. Operational flexibility within the project boundary is provided by battery operation subject to state-of-charge dynamics and mutually exclusive charge/discharge, together with bounded grid exchanges at the PCC. Ramping constraints for conventional generators are not modelled because upstream dispatch outside the project boundary is treated as exogenous. Battery ramp-rate limits are not imposed because battery response times are much faster than the hourly time step used in this study. A conventional reserve-margin constraint is not included because adequacy is enforced directly at the project boundary via the 8760-h balance under the PCC exchange limit, together with substation sizing and N-1 security.

2.4. Simulation scenarios

To examine how coordinated planning of renewables, storage, and grid interactions responds in both investment and operations under different levels of clean electricity stringency, we define four renewable energy supply target scenarios: 30%, 50%, 70%, and 95%. These targets are specified exogenously as scenario inputs. The 30% target is policy-driven, aligned with China's national baseline requirement for direct green power connection projects (summarised in SI Table S1), whereas 50%, 70%, and 95% are scenario-based stress tests to examine feasibility–cost transitions as targets tighten toward near self-sufficiency. This gradient spans from moderate green power procurement to near quasi off grid deep decarbonisation constraints, enabling us to reveal the structural transition from grid purchase dominated supply toward locally provided wind and solar PV supported by storage-based balancing. It also allows us to characterise the tradeoffs between CAPEX and OPEX and the associated changes in interconnection capacity sizing. Scenario definitions are summarised in Table 3.

Under this setup, the 30% target serves as a baseline representing high reliance on grid supply. The 50% target is used to test whether, as

the RE constraint tightens, private wire wind, solar PV, and storage enter a scale up regime and how the buy–sell strategy adjusts. The 70% target quantifies the incremental costs and binding bottlenecks at high RE shares. The 95% target acts as a stress test to identify which combinations of constraints and drivers trigger sharp cost escalation.

Across these four scenarios, we systematically compare the optimal DC siting and the spatial pattern of wind and solar PV deployment, battery power and energy sizing and implied duration, PCC capacity sizing, and changes in total cost and cost composition. The results provide quantitative guidance for siting and capacity configuration of private wire supplied DCs under direct green power connection schemes.

3. Results

The results are organised to mirror the workflow. Section 3.1 maps developable land for utility-scale wind and solar PV in Qinghai after applying topographic, land-use, and ecological screening, and summarises the associated annual technical generation potential. Section 3.2 characterizes prefecture-level DC electricity demand profiles together with climate-driven PUE variability, showing how cooling conditions translate into differences in total facility load under a consistent IT sizing assumption. Section 3.3 then reports MILP co-optimisation outcomes under alternative renewable supply targets, including private-wire hub siting and supply allocation patterns, storage sizing, cost breakdowns and trade-offs, and the implied grid electricity purchase displacement.

3.1. Developable land and technical potential for wind and solar PV

We quantify developable land and annual technical generation potential for wind and solar PV in Qinghai. A deployability mask is first constructed by integrating land-use, ecological, and engineering constraints, and resource-based outputs are then aggregated over feasible cells to obtain grid-level annual technical potential. Fig. 5 maps the developable areas, while Figs. 6 and 7 report the annual technical generation potential for wind and PV across those cells. Overall, developable cells concentrate in northern and northwestern Qinghai and form contiguous zones suitable for utility-scale deployment. In contrast, areas such as Yushu are strongly constrained by ecological protection and land-use restrictions, leaving only fragmented developable patches that offer limited contiguous sites for utility-scale deployment.

Within the developable area, the annual technical generation potential of wind exhibits a pronounced hotspot pattern (Fig. 6). High potential cells cluster in bands and patches along the dominant wind corridor from northern to northwestern Qinghai, indicating strong spatial selectivity for wind development. Haixi Tibetan and Mongol Autonomous Prefecture stands out, with concentrated and contiguous high potential zones. In contrast, areas around Haidong and Xining are dominated by medium to low values or fragmented patches, and Golog Tibetan Autonomous Prefecture is also largely characterised by medium to low potential.

By comparison, the annual technical generation potential of solar PV is smoother and more spatially continuous (Fig. 7). Northern Qinghai forms a broad belt of medium to high potential, reflecting a more stable provincial scale solar resource and an areal deployment pattern. High potential zones are concentrated in Haixi, Haibei, and Hainan, forming a relatively consistent northern band. Although Haidong and Xining in the east include a nontrivial developable footprint, their intensity is generally lower than the northern high potential belt, while the southern developable areas are overall weaker. Taken together, both wind and PV high potential cells concentrate in northern to northwestern Qinghai; wind requires finer hotspot selection, whereas PV offers a more continuous and scalable deployment basis.

3.2. Urban data center demand and climate driven PUE variability

On the demand side, to characterise the spatial and temporal patterns of DC loads across cities, we aggregate candidate grid level loads within each prefecture level city or autonomous prefecture. For each grid cell, we integrate the hourly load over the year to obtain annual electricity consumption, and map it onto the provincial grid to produce the spatial distribution of annual DC demand in Qinghai (Fig. 8). At the city scale, using the aggregated 8760 h load series for each city, we construct load duration curves (Fig. 9) and average daily load profiles (Fig. 10) to compare demand levels and variability across regional DC clusters under a consistent IT capacity assumption.

Fig. 8 shows a clear spatial gradient in annual DC electricity consumption at the grid level. Annual demand per grid cell is generally around 2.5×10^6 MWh, yet spatial heterogeneity within the province is pronounced. A higher load corridor emerges across the north from west to east, and grid cells near Xining and Haidong in the east rise further to form the most concentrated high load cluster in Qinghai. In contrast, southern areas and peripheral northern grid cells mostly fall in the low to medium range. This pattern suggests that, under the unified sizing assumption, cooling energy differences driven by climate and PUE variation are not extreme, but the relatively milder conditions in northern and western Qinghai still lead to slightly higher annual loads.

The load duration curves based on city aggregated demand (Fig. 9) indicate clear interregional differences. Haixi Tibetan and Mongol Autonomous Prefecture exhibits the highest peak load, exceeding other regions markedly during the few most heavily loaded hours of the year, suggesting a more concentrated set of extreme load periods and a more pronounced peak. Haidong and Haibei show consistently higher curves, implying a larger share of high load hours. In contrast, Huangnan, Hainan, and Golog display flatter curves with smaller peak to trough spreads, with demand remaining slightly below the provincial average for most of the year.

Fig. 10 shows the average diurnal patterns. Across all cities, daily load profiles are relatively flat, with limited intraday variation, consistent with high-utilisation DCs operating close to baseload demand. Because the IT sizing is held constant across prefectures, inter-regional differences in total facility load are primarily driven by climate-induced PUE differences in our model. The modest diurnal shape, including the relative trough around midday, reflects the diurnal cycle of wet-bulb temperature, which modulates cooling demand and, in turn, PUE, with cooler periods enabling more economiser/free-cooling operation. Against this backdrop, Haidong and Haibei exhibit noticeably higher average daily demand than other regions, with a modest peak from early morning to late morning, a slight dip in the afternoon, and a renewed rise toward evening. Xining and Haixi also show somewhat higher profiles than the rest of the province, and follow a similar shape, with “higher in the early morning and a relative trough around midday”. In contrast, southern regions such as Huangnan, Hainan, Golog, and Yushu have lower overall demand and gentler intraday fluctuations,

Table 3
Simulation scenarios and interpretations.

Scenario ID	Scenario name	RE supply target	Interpretation
S1	Baseline	30%	Current-policy case; used to obtain the least-cost feasible solution and serve as the benchmark.
S2	Moderate transition	50%	Medium-stringency case; represents system configuration and operational responses under mid-term renewable electricity targets.
S3	High renewables	70%	High-stringency case; requires a substantial increase in local wind and solar self-supply and stronger balancing from energy storage.
S4	Extreme constraint	95%	Stress-test case; characterizes the feasibility boundary and the upper bound of system cost under near off-grid, near-fully renewable operation.

reinforcing a “low level, low variability” demand signature.

Taken together, Figs. 8–10 indicate that, under a uniform DC sizing assumption, northern and eastern Qinghai correspond to higher annual consumption, higher peaks, and a larger share of high load hours. For subsequent planning of private wire wind–solar–storage hubs, this implies that Haixi, Haidong, Haibei, and Xining require particular attention to peak hour supply adequacy and PCC headroom to ensure feasible operation under hourly balance and reliability constraints.

3.3. Private wire hub siting and investment operation tradeoffs

Under the private wire supply framework, this section examines the coordinated decisions on **DC** hub siting and the associated wind and solar PV investment portfolio. Using the MILP optimal solutions under different renewable supply targets, we extract the selected **DC** hub grid cell in each prefecture, together with the wind and solar PV build locations and capacities coupled to the hub via dedicated lines. We then represent these results at the provincial scale as spatial combinations of “hub locations” and “supply locations” (Figs. 11–14). The full coordinates (latitude and longitude) of the selected DC hubs and the associated PV and wind sites under each target are provided in Appendix A (Table A1). This analysis reveals how, as the renewable constraint tightens, the system reconfigures siting and capacities under joint constraints of resource endowments, dedicated line distances, and PCC access costs, thereby forming implementable layouts for direct green power connections to **DCs**. To test the robustness of siting outcomes under prospective declines in renewable equipment costs, we conduct a sensitivity analysis for the policy-baseline 30% case by synchronously scaling wind and PV CAPEX (×1.0, ×0.8, ×0.5). Table S4 summarises whether the optimal DC hub grid cell and selected wind/PV grid cells change across these cost assumptions.

Under the 30% target (Fig. 11), the optimal solution follows a pattern of “prioritising nearby interconnection for the hub, using low barrier PV to meet the RE requirement, and excluding wind”. DC hubs are located closer to load centres and areas with favourable corridor conditions, with clustering around Xining and Haidong. On the supply side, only solar PV is built, and capacity differences across prefectures are small, at roughly 334–378 MW. At this stage, a large share of demand can still be met by grid purchases, so the model prefers PV, which is flexible to site and easier to couple in the hub's near field, rather than taking on the additional siting and dedicated line costs associated with wind.

When the target increases to 50% (Fig. 12), the adjustment occurs primarily on the supply side, shifting “from marginal top up to capacity expansion”, while hub siting continues to be driven by minimising interconnection costs. DC locations largely remain within candidate grid cells that are more favourable for access and coupling within each administrative unit, and density remains higher around the eastern load centre. This indicates that the model does not rely on large hub relocation to reduce costs, but instead satisfies the tighter constraint mainly through supply expansion. Solar PV remains the backbone, but installed capacity increases substantially and becomes more differentiated, reaching about 437–683 MW. Wind also enters the optimum, yet it is concentrated in Haixi (about 355 MW) rather than being deployed province wide. This suggests that, at this stage, wind acts as a marginal resource: in the best wind zones, its higher energy yield per unit capacity substitutes for part of the PV expansion, moderating the increase in system cost.

Under the 70% target (Fig. 13), this structural division becomes more pronounced. Hub locations change only marginally, but within each jurisdiction they converge more clearly toward candidate grid cells with better corridor conditions or construction suitability. The supply side shifts more substantially. Except for Haixi, solar PV capacity in most regions rises to roughly 860–950 MW, indicating that as the headroom for grid purchases shrinks further, the system increasingly relies on the

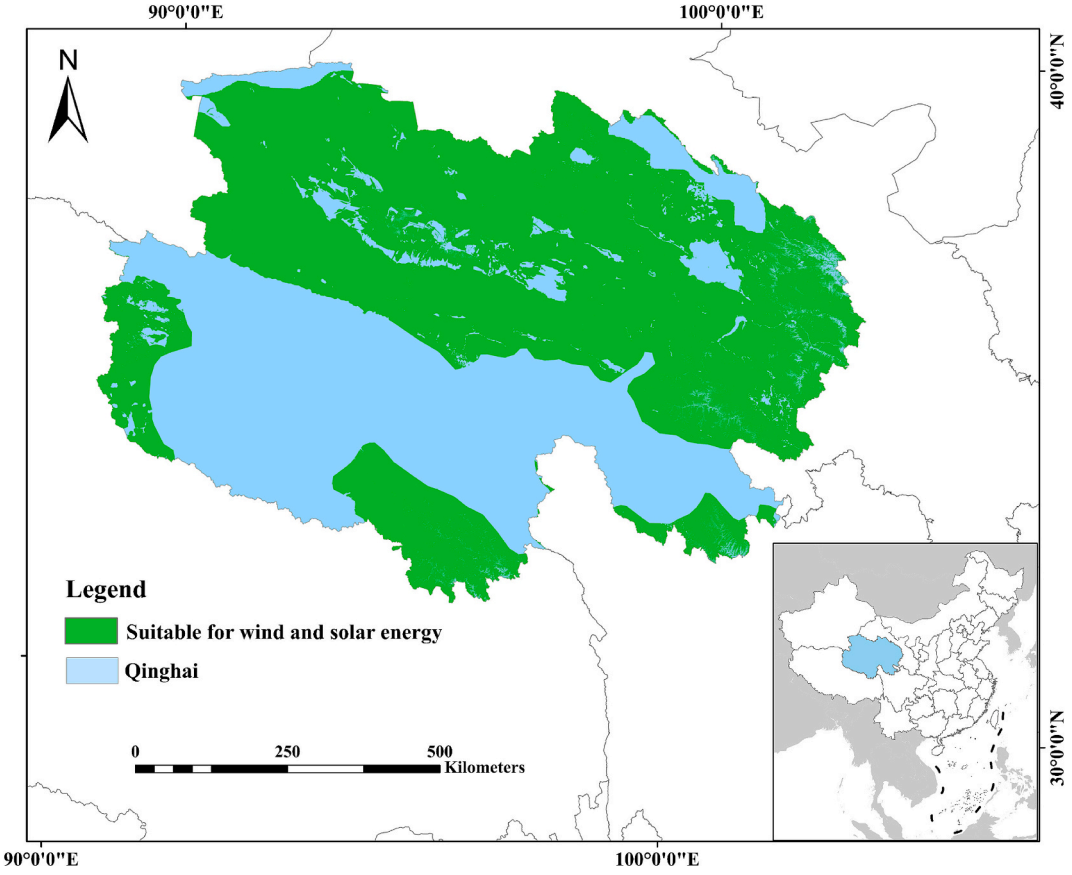


Fig. 5. Developable resource suitability for wind and solar deployment in Qinghai.

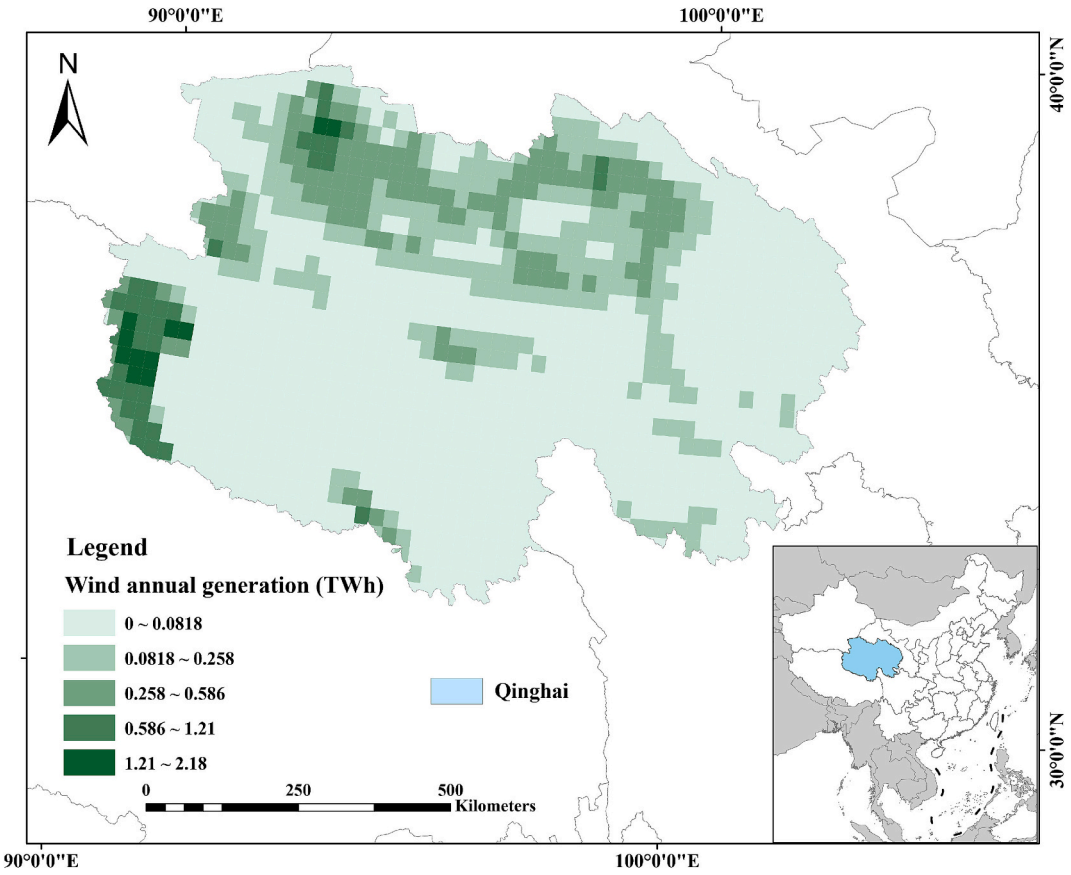


Fig. 6. Annual wind technical potential in developable areas.

scalability of PV as the main backstop. In Haixi, PV capacity is instead lower (about 601 MW), while it becomes the only region that deploys wind (about 392 MW). This pattern is not simply "better resources imply more capacity", but rather reflects an active trade off between capacity efficiency and dedicated line costs under stringent constraints. Introducing wind in the best wind corridor yields more effective renewable energy with less installed capacity, reduces the redundant PV build required under a PV only strategy, and helps mitigate potential curtailment risk.

When the target is further raised to 95% (Fig. 14), the system enters a near self sufficiency regime, and the siting logic shifts from cost minimisation toward feasibility and coupling. Hub locations no longer prioritise nearby interconnection alone, but also emphasise reachability to supply sites and hourly feasible operation under stringent self consumption and export limits. The supply side exhibits a clearer hierarchy. Solar PV delivers most of the incremental requirement and forms a multi node support pattern, while wind contracts to a strategic supplement, remaining only in Haixi at a small scale (about 173 MW). This indicates that, near quasi off grid operation, wind is not deployed broadly but is used only where resource and access conditions are advantageous and where it can effectively fill critical deficit hours, avoiding amplified line and construction costs in non optimal areas. Overall, the four stages follow a consistent progression: PV based top up at low targets, PV expansion with wind introduced in the best wind corridor at medium targets, PV as a scalable backstop with wind substituting in the most advantageous areas at high targets, and a near self-sufficiency stage driven by coupling feasibility with wind reduced to local supplementation.

When the target is further raised to 95% (Fig. 14), the system enters a near self sufficiency regime, and the siting logic shifts from cost minimisation toward feasibility and coupling. Hub locations no longer prioritise nearby interconnection alone, but also emphasise reachability to supply sites and hourly feasible operation under stringent self consumption and export limits. The supply side exhibits a clearer hierarchy. Solar PV delivers most of the incremental requirement and forms a multi node support pattern, while wind contracts to a strategic supplement, remaining only in Haixi at a small scale (about 173 MW). This indicates that, near quasi off grid operation, wind is not deployed broadly but is used only where resource and access conditions are advantageous and where it can effectively fill critical deficit hours, avoiding amplified line and construction costs in non-optimal areas. Overall, the four stages follow a consistent progression: PV based top up at low targets, PV expansion with wind introduced in the best wind corridor at medium targets, PV as a scalable backstop with wind substituting in the most advantageous areas at high targets, and a near self-sufficiency stage driven by coupling feasibility with wind reduced to local supplementation.

Table 4 summarises the optimal storage configurations across prefectures under the four target scenarios, reported in terms of storage power, storage energy, and implied duration. Storage is not merely an add-on, but a key mediator that absorbs the temporal variability of wind and PV, covers deficit hours, and increases the effective utilisation of renewable energy under self-consumption and export constraints. It therefore directly determines feasibility in high target cases.

Results show that no region builds storage in the 30% scenario, implying that under loose constraints the system can rely on grid exchanges for balancing and storage provides insufficient marginal value to justify its capital cost. Once the target increases to 50%, most regions except Haixi deploy short duration storage of about 300 MW and 600 MWh (2 h), indicating that under moderate constraints the system first adopts high power, short duration storage to manage intra day fluctuations and reduce reliance on grid purchases. At the 70% target, storage scales up markedly and diverges along two pathways. One pathway continues to expand 2 h systems, with Golog and Haibei reaching

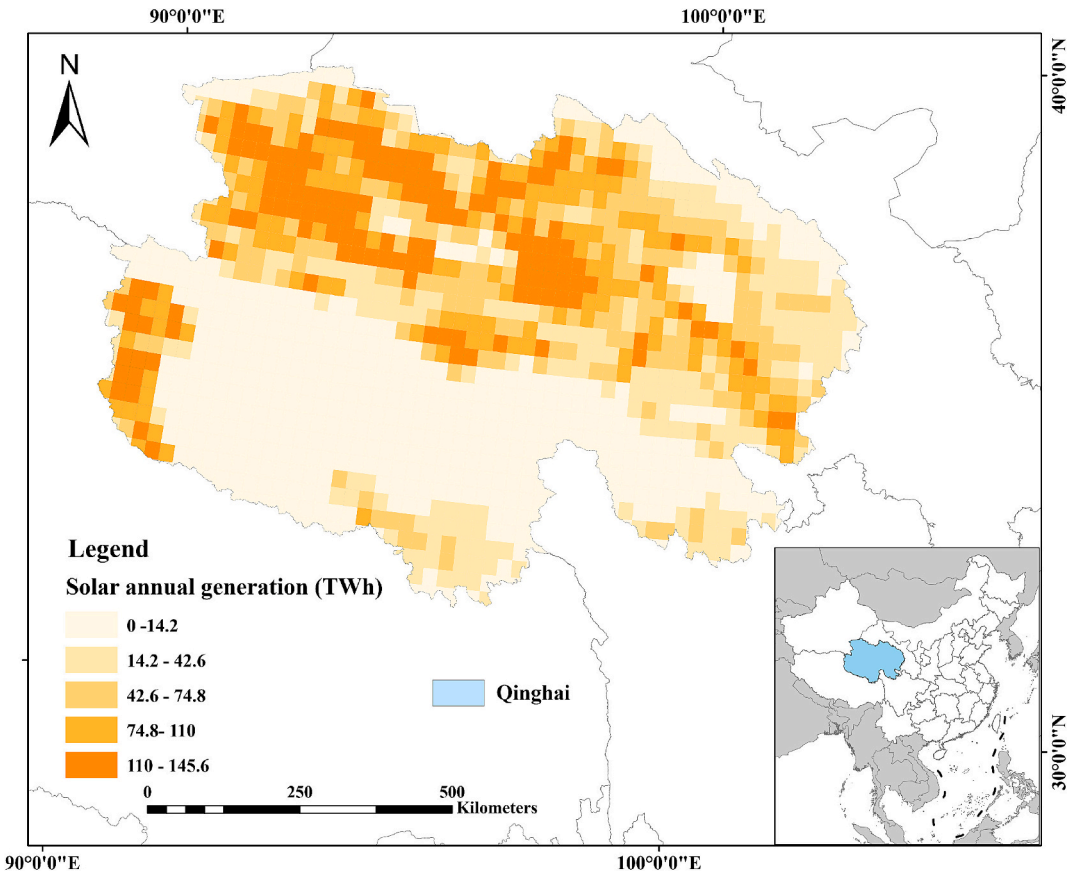


Fig. 7. Annual solar PV technical potential in developable areas.

roughly 800 MW and 1600 MWh. The other extends toward 4 h duration, with Xining, Haidong, Yushu, and Huangnan favouring around 450 MW and 1800 MWh. This divergence reflects the need, under tighter constraints, to move beyond short term regulation toward stronger intertemporal energy shifting capability to mitigate longer time scale supply–demand mismatches.

Under the 95% near self-sufficiency scenario, storage becomes the pivotal asset for maintaining feasibility, and intercity heterogeneity widens markedly, reflecting strategy differentiation jointly driven by resource conditions and supply portfolio choices. One group opts for high power, short duration configurations. For example, Haidong deploys about 1.61 GW and 3.22 GWh (2 h), and Huangnan about 1.58 GW and 3.16 GWh (2 h), prioritising rapid support during the most severe deficit hours. Another group favours moderate power with longer duration. Hainan, Haixi, and Xining adopt 6 h storage, indicating that extending duration can be more cost effective than simply increasing instantaneous power. Golog, Haibei, and Yushu mainly fall in the 4 h range, consistent with a systematic shift from short to medium duration storage as constraints tighten.

We further assess the cost implications of the private wire configuration and how cost structures evolve across renewable targets. Table 5 reports total system costs for each prefecture under a pure grid only baseline and under the 30–95% targets. Under the grid only baseline, costs are similar across regions at about $0.68–$0.69 billion. Once private wire is introduced and the RE constraint is tightened, total costs increased approximately monotonically with target stringency: $1.76–$1.96 billion at 30%, $3.23–$3.82 billion at 50%, $5.28–$6.18 billion at 70%, and $8.56–$9.56 billion at 95%. At the same time, cross city cost dispersion expands substantially as targets tighten. At 95%, Haidong and Huangnan sit at the upper end, whereas Haixi remains relatively lower. This pattern indicates that near self-sufficiency, satisfying stringent share targets and hourly balance requires higher levels of local renewable supply and storage redundancy to secure feasibility, which increases capital intensive investments and amplifies cost stratification driven by heterogeneous resource endowments.

Fig. 15 further illustrates the mechanism driving the shift in cost structure. At the 30% target, operating costs are nearly identical across regions at about $0.49 billion, indicating that grid purchases and sales remain the primary means of balancing. In contrast, investment costs reflect only the minimum incremental build needed to satisfy the constraint. As the target increases to 50% and 70%, operating costs fall to roughly $0.33–$0.35 billion and $0.19–$0.23 billion, respectively, consistent with rising shares of local wind and solar PV supply and storage-based balancing. Grid transactions are reduced to a boundary adjustment role, while capital expenditures increase rapidly and become the dominant component of total cost. A more detailed decomposition of CAPEX into wind/PV, storage, dedicated lines/interconnection, and substation components is provided in Fig. S1.

In the 95% near self-sufficiency regime, operating costs are compressed further to about $0.03–$0.06 billion, and total costs are almost entirely determined by investment, such that city level differences in resources and strategies translate into persistent cost stratification. Haidong illustrates this pattern: its total cost reaches $9.56 billion at 95%, consistent with the high power, short duration storage configuration in Table 4 of about 1.61 GW and 3.22 GWh (2 h), reflecting stronger power adequacy pressure during deficit hours. Huangnan shows a similar high-cost outcome ($9.43 billion at 95%) with an analogous storage pathway. By contrast, Haixi remains consistently lower under medium to high targets ($8.56 billion at 95%), aligning with its ability to exploit complementary resources in superior wind zones and reduce redundant capacity needs. This highlights that resource complementarity can sustain hourly feasibility under stringent share constraints at a lower capital cost.

From the abatement perspective, Fig. 16 reports the annual grid

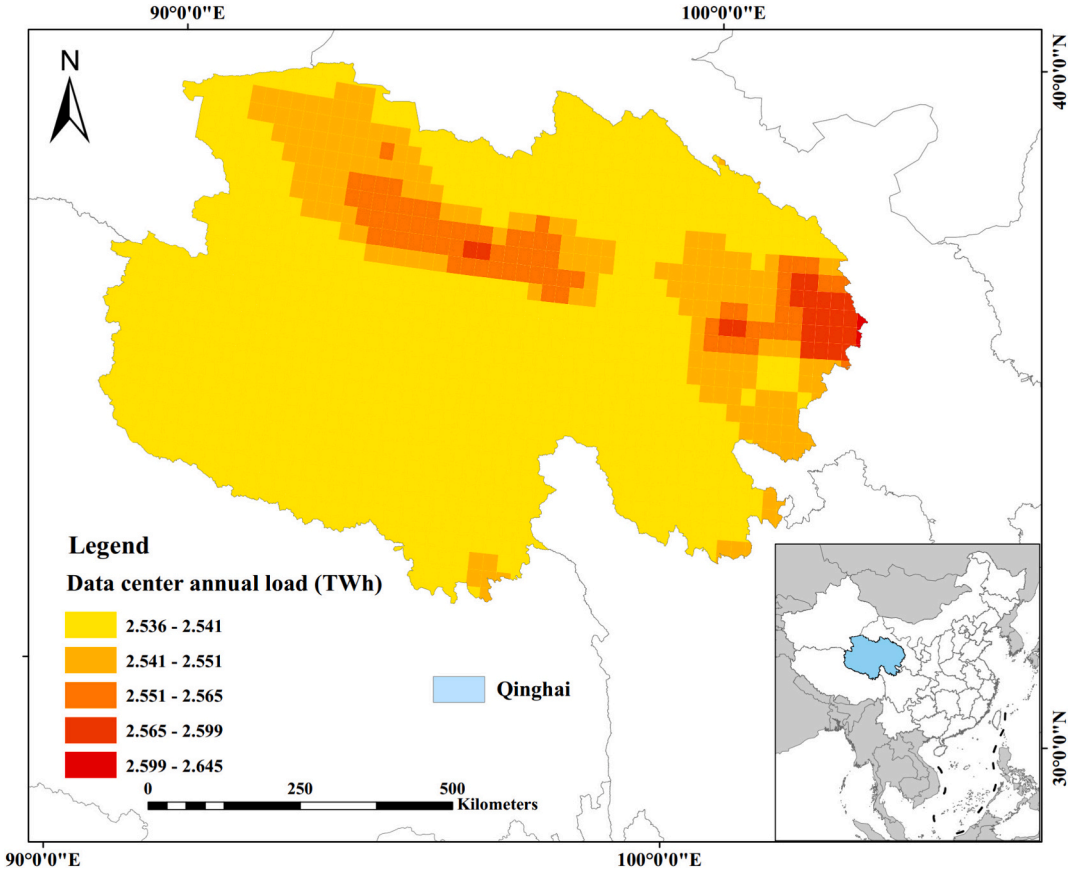


Fig. 8. Spatial distribution of annual data center electricity load in Qinghai.

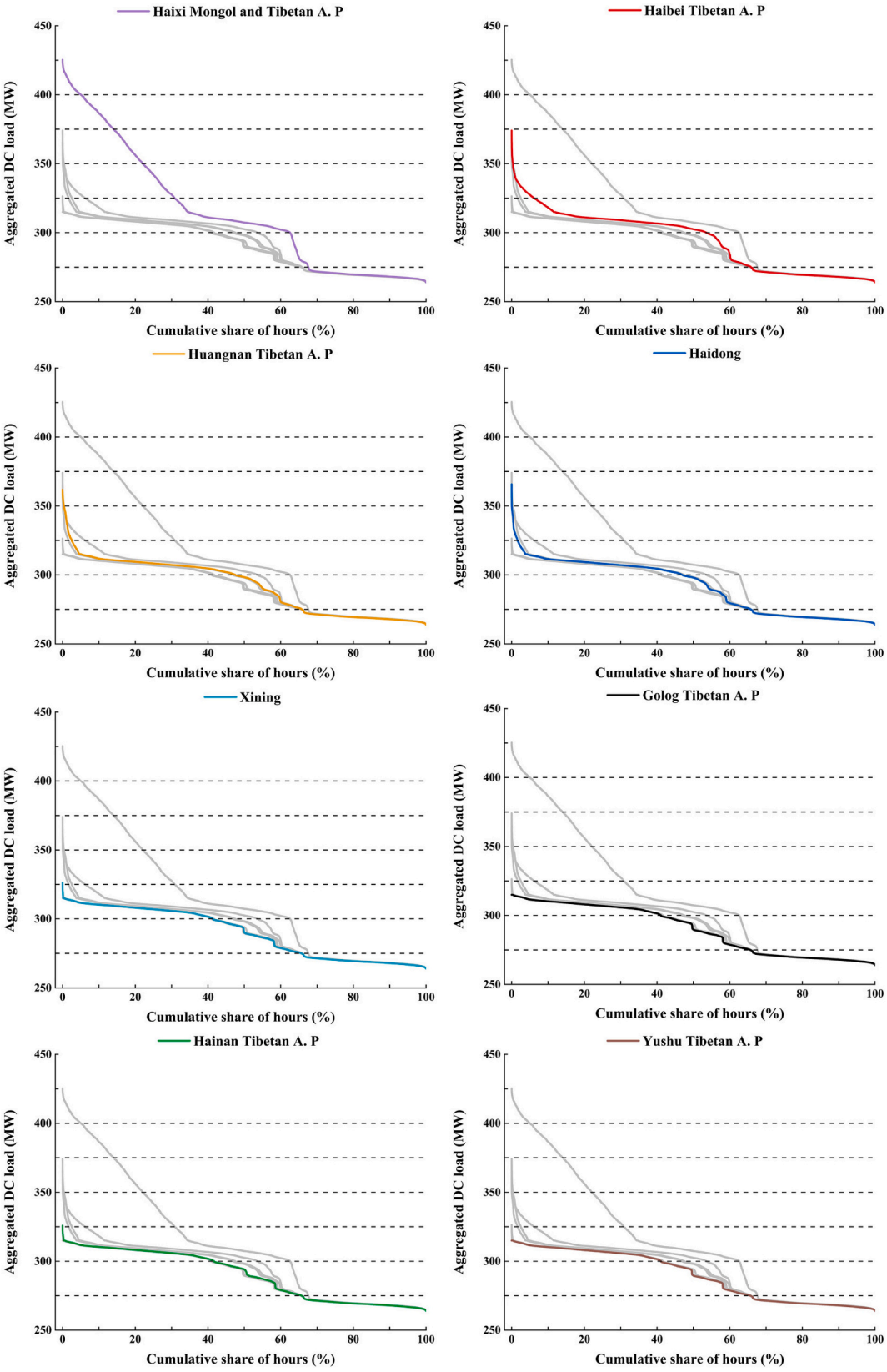


Fig. 9. Prefecture level data center load duration curves in Qinghai. Each panel highlights one city in colour, while the grey curves show the corresponding load-duration curves for the other prefectures for comparison.

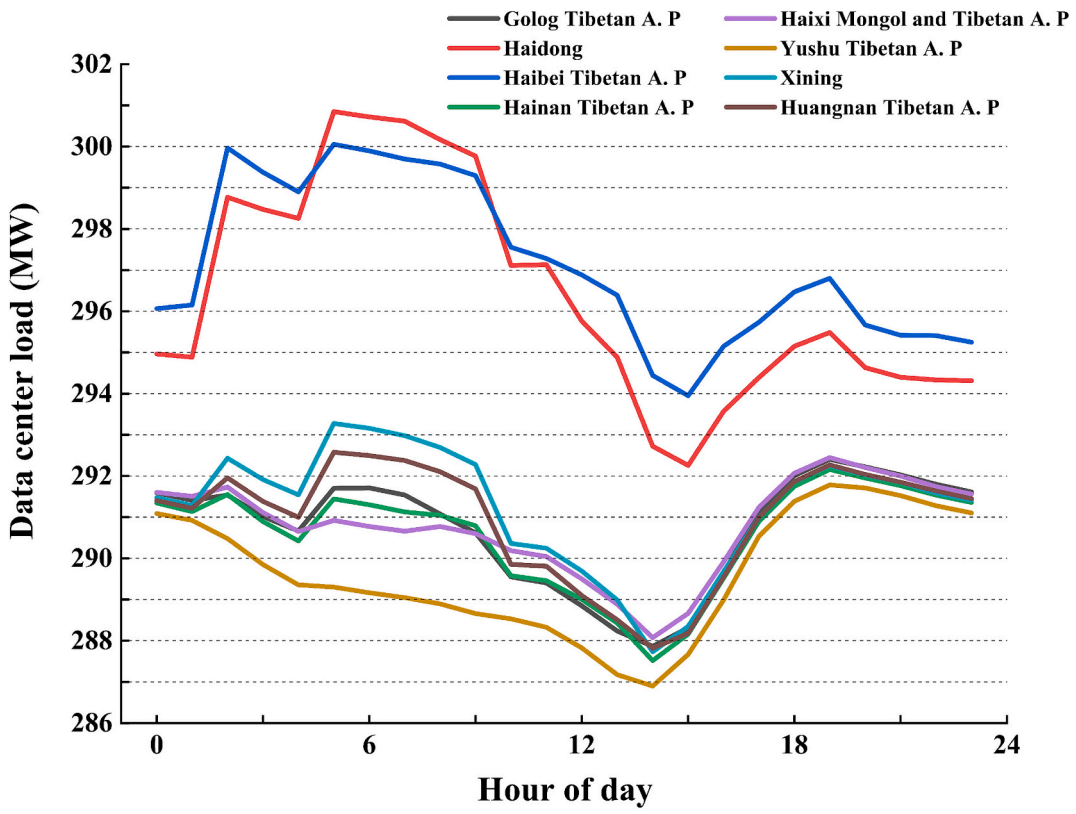


Fig. 10. Mean diurnal profile of data center load by prefecture.

electricity purchase displacement achieved by the private wire wind, solar PV, and storage system under different targets, defined as the reduction in grid purchases relative to the grid only baseline for the same DC demand. As the target tightens from 30% to 95%, the annual grid purchase displacement per city increases from about 0.76 TWh to roughly 2.17–2.26 TWh, meaning an additional 1.41–1.50 TWh yr^{-1} of DC electricity demand per city is supplied by the matched wind-solar-storage system rather than purchased from the public grid. Using Qinghai's provincial average grid emission factor of 0.1796 $kgCO_2/kWh$ [66], the implied annual emissions reduction per city increases from about 0.14 Mt. CO_2 at 30% to about 0.39–0.41 Mt. CO_2 at 95%. These results indicate that coordinated private wire planning not only shifts cost structure, but also yields a transparent and auditable scale of grid electricity displacement and associated emissions reductions as targets tighten.

4. Discussion

4.1. Mechanism analysis and synthesized findings

Our results reveal two robust patterns. As the renewable penetration target tightens, grid electricity purchase displacement within the project boundary increases markedly, while the cost structure shifts rapidly from being operating-driven to capital-driven, leading to pronounced inter-city differentiation at high targets. When the renewable penetration target increases from 30% to 95%, annual grid electricity purchase displacement rises from about 0.76 TWh to roughly 2.17–2.26 TWh per city. This indicates that private-wire supply can deliver auditable, scalable physical displacement.

The root cause of inter-city cost divergence is not whether annual energy is sufficient, but whether hour-by-hour deliverability is feasible. Under low-to-mid renewable penetration targets, the system can still rely on a degree of grid trading for balancing, requiring only limited incremental wind–solar build-out; operating costs therefore account for a larger share of total cost, and city-to-city differences remain modest. As the target approaches near self-sufficiency, the PCC exchange limit and the private-wire deliverability boundary jointly compress the balancing flexibility available from the public grid. Hourly balance and reliability constraints then become dominant, and feasibility requires more local wind/PV and storage redundancy. As a result, capital expenditures become the primary determinant of total cost, and inter-city differences widen substantially.

The joint sizing of storage power and energy explains the divergence between “high-cost” and “low-cost” cities under stringent targets. When binding bottlenecks concentrate in a small set of deficit hours, optimal solutions tend to favour higher storage power with shorter duration to satisfy power adequacy. When constraints are driven more by inter-temporal energy shifting, longer-duration energy capacity becomes more valuable. Cities with stronger renewable resource complementarity can reduce redundant build-out and balancing asset requirements, thereby sustaining lower capital cost and more robust feasibility under medium-to-high targets.

4.2. Benchmarking with existing literature and practical validation

Our results show that, as the clean-electricity target tightens, the system shifts from a low-cost regime in which grid trading provides the main balancing mechanism to a capital-intensive regime in which

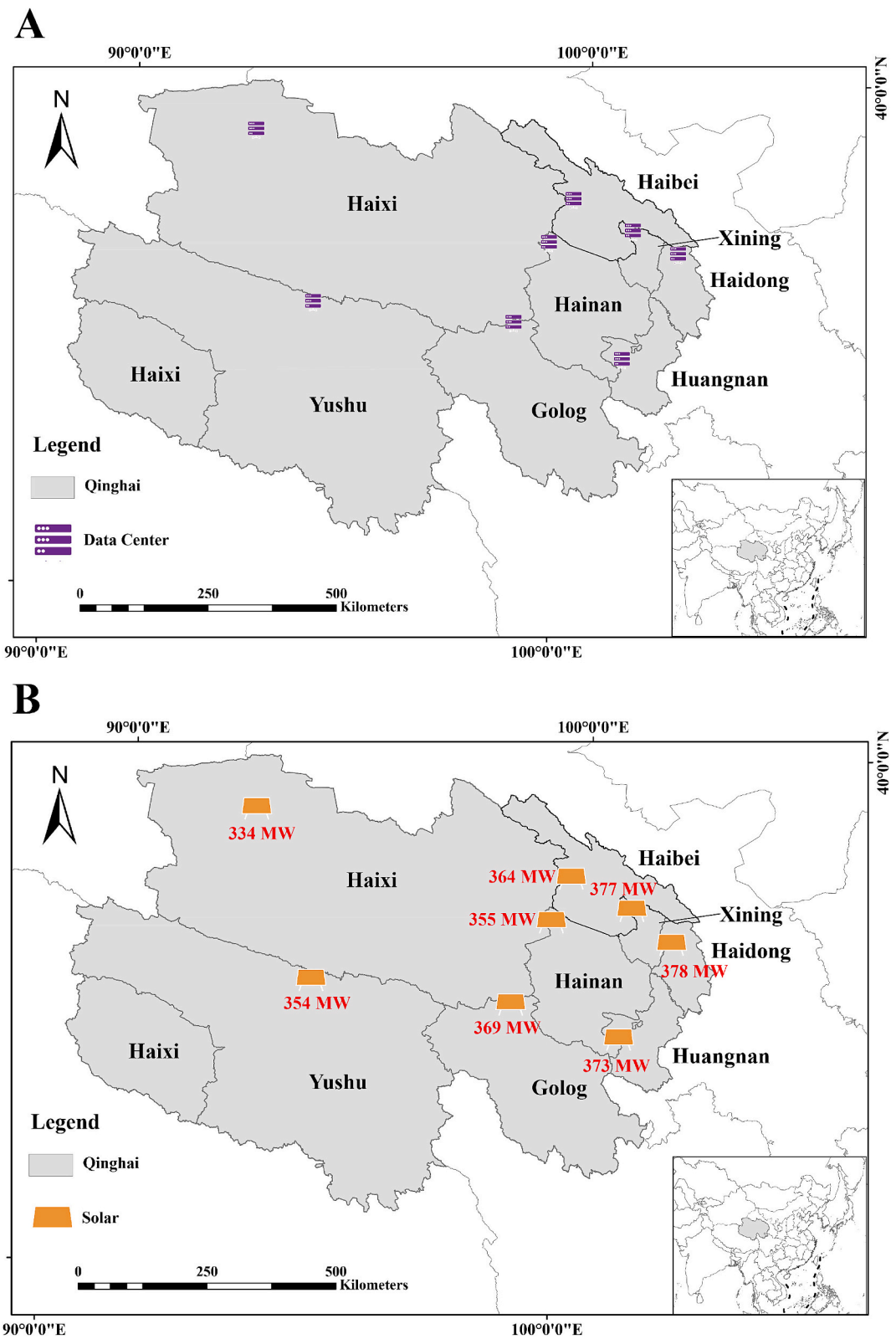


Fig. 11. Optimal siting in Qinghai under the 30% renewable target. (A) Data center (B) Solar PV.

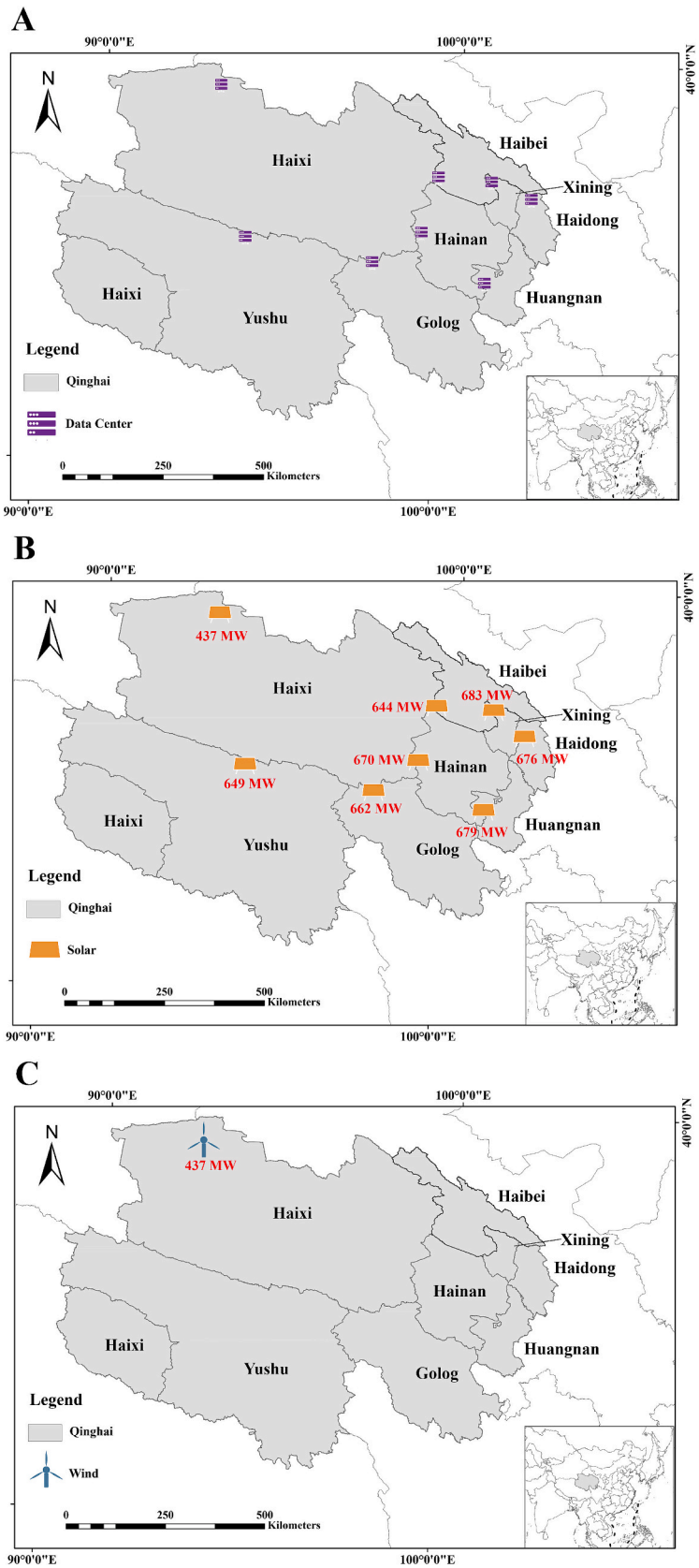


Fig. 12. Optimal siting in Qinghai under the 50% renewable target. (A) Data center (B) Solar PV (C) Wind.

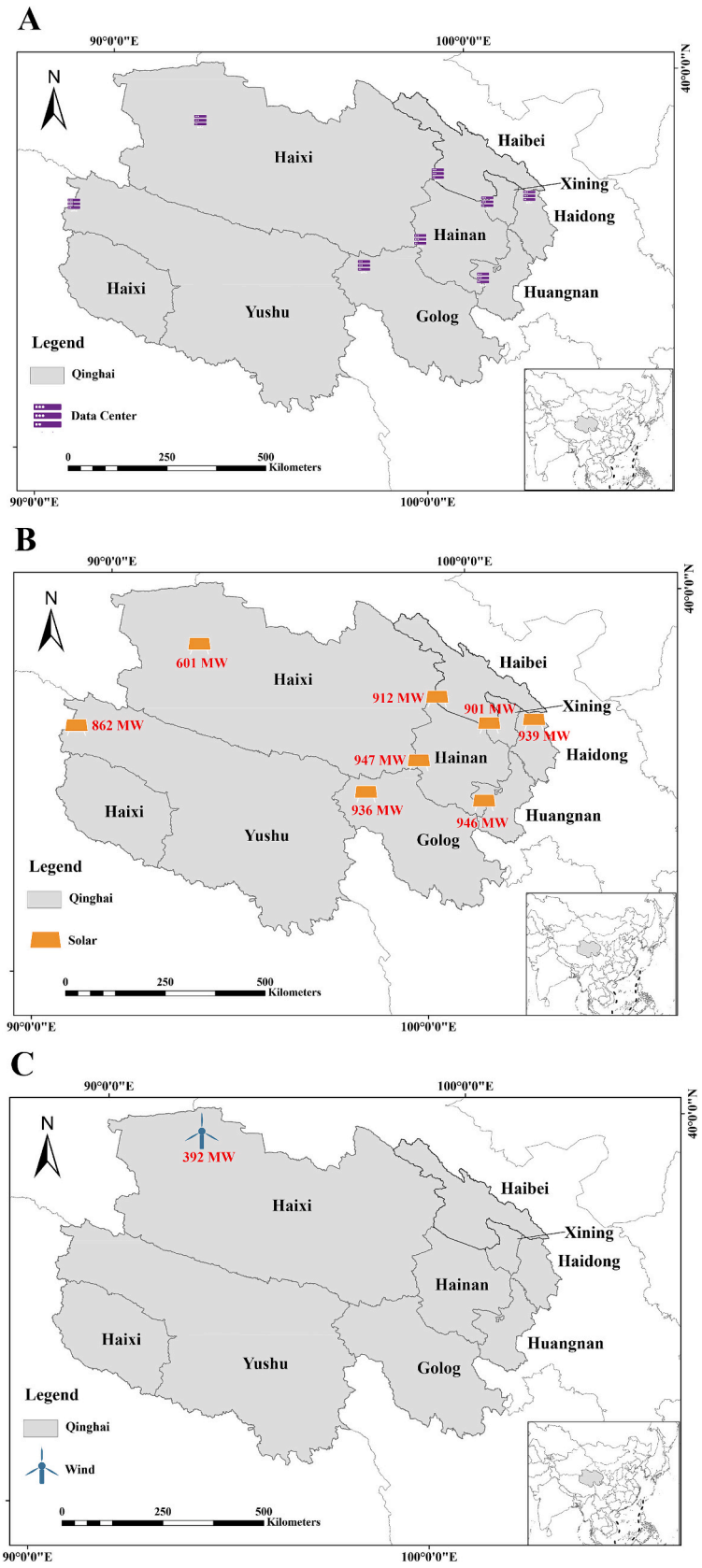


Fig. 13. Optimal siting in Qinghai under the 70% renewable target. (A) Data center (B) Solar PV (C) Wind.

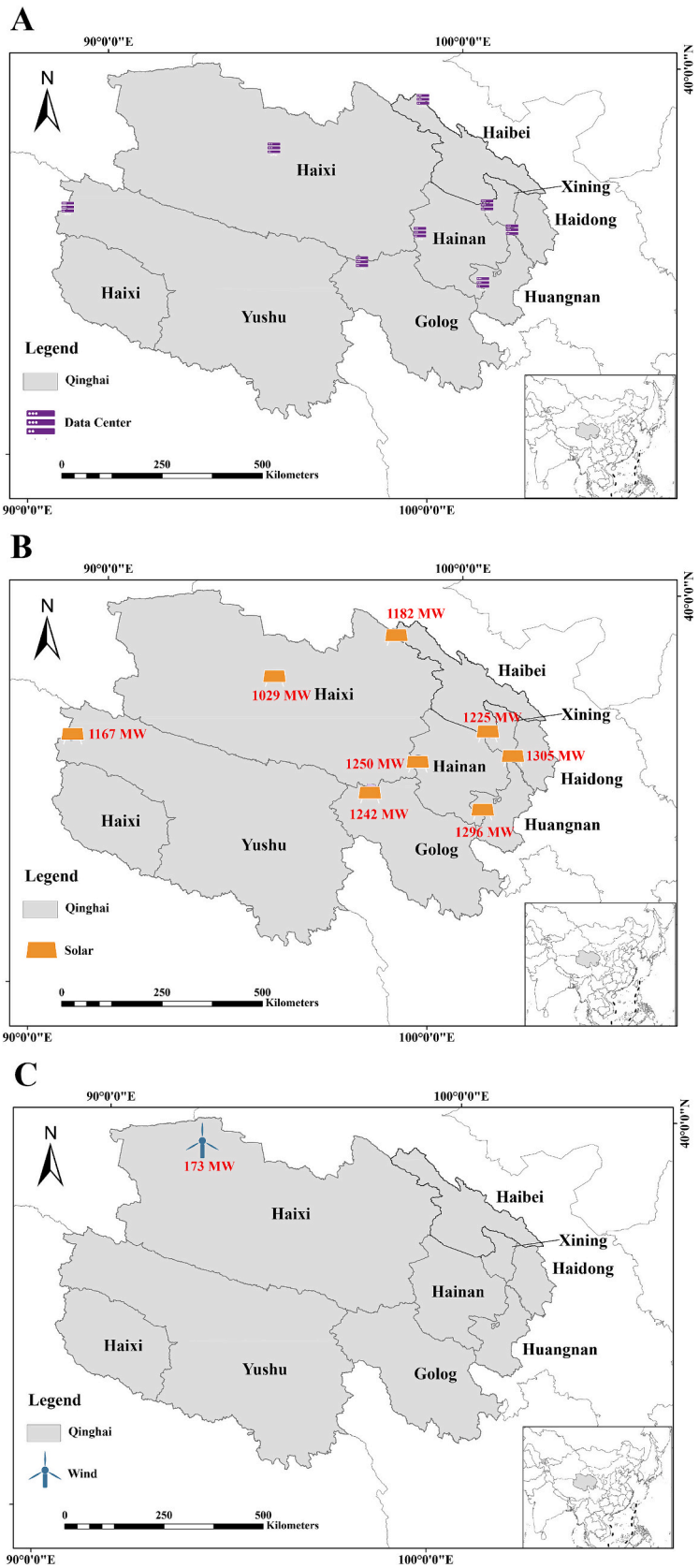


Fig. 14. Optimal siting in Qinghai under the 95% renewable target. (A) Data center (B) Solar PV (C) Wind.

Table 4
Optimal battery power, energy, and implied duration under different renewable targets.

City	Renewable target	Storage power (MW)	Storage energy (MWh)	Storage duration (h)
Golog Tibetan A. P	30%	0	0	–
Golog Tibetan A. P	50%	326	651	2
Golog Tibetan A. P	70%	820	1641	2
Golog Tibetan A. P	95%	819	3275	4
Haidong	30%	0	0	–
Haidong	50%	333	666	2
Haidong	70%	462	1847	4
Haidong	95%	1611	3223	2
Haibei Tibetan A. P	30%	0	0	–
Haibei Tibetan A. P	50%	326	653	2
Haibei Tibetan A. P	70%	824	1648	2
Haibei Tibetan A. P	95%	827	3309	4
Hainan Tibetan A. P	30%	0	0	–
Hainan Tibetan A. P	50%	317	635	2
Hainan Tibetan A. P	70%	816	1632	2
Hainan Tibetan A. P	95%	573	3435	6
Haixi Mongol and Tibetan A. P	30%	0	0	–
Haixi Mongol and Tibetan A. P	50%	0	0	–
Haixi Mongol and Tibetan A. P	70%	436	872	2
Haixi Mongol and Tibetan A. P	95%	502	3013	6
Yushu Tibetan A. P	30%	0	0	–
Yushu Tibetan A. P	50%	309	618	2
Yushu Tibetan A. P	70%	458	1830	4
Yushu Tibetan A. P	95%	818	3274	4
Xining	30%	0	0	–
Xining	50%	328	657	2
Xining	70%	458	1830	4
Xining	95%	575	3448	6
Huangnan Tibetan A. P	30%	0	0	–
Huangnan Tibetan A. P	50%	327	653	2
Huangnan Tibetan A. P	70%	457	1829	4
Huangnan Tibetan A. P	95%	1582	3164	2

Note: Storage duration is calculated ex post as storage energy divided by storage power (E_b/P_b) and rounded to the nearest integer hour for presentation.

Table 5
Total system cost by city under different renewable targets.

City	Grid only baseline (billion $)	Renewable target (billion $)			
		30%	50%	70%	95%
Golog Tibetan A. P	0.68	1.87	3.67	6.12	9.07
Haidong	0.69	1.96	3.82	6.18	9.56
Haibei Tibetan A. P	0.68	1.88	3.6	5.92	9.19
Hainan Tibetan A. P	0.68	1.82	3.65	5.99	9.04
Haixi Mongol and Tibetan A. P	0.69	1.76	3.23	5.28	8.56
Yushu Tibetan A. P	0.68	1.82	3.56	5.9	8.78
Xining	0.68	1.9	3.73	5.91	8.99
Huangnan Tibetan A. P	0.69	1.89	3.71	6.03	9.43

hourly feasibility is increasingly secured by local wind/PV and storage. This pattern is broadly consistent with recent studies on hourly matching and physically grounded DC clean-supply systems, which likewise find that stricter temporal and physical delivery requirements increase dependence on storage and local clean capacity. Our contribution is to explicitly show this transition within private-wire deliverability and PCC boundaries, and to link it to the observed shift in cost structure and the emergence of inter-city divergence.

An operating green power direct-connection computing project provides an engineering benchmark for the low- to mid-target regime identified in our results. Public information indicates that the project deploys approximately 200 MW of wind, 100 MW of solar PV, and 45 MW storage (4 h), achieving about 0.848 TWh yr^{-1} of self-consumed green electricity and a renewable substitution rate of roughly 38.74% [67,68]. This scale and operating profile are consistent with our 30–50% target cases, in which self-consumption dominates, export remains limited, and grid transactions mainly play a boundary-adjustment role under deliverability constraints. Differences in absolute capacity levels across projects are expected and are primarily driven by DC scale, site conditions, and deliverability boundary settings, but they do not alter the mechanism-level conclusions.

4.3. Policy implications and industrial relevance

The central policy implication is that green power direct connection should not stop at annual share compliance, but should progressively incorporate engineering-grounded, hour-by-hour deliverability requirements, and link them to auditable abatement outcomes. Across candidate-city cases, annual grid electricity purchase displacement increases from about 0.76 TWh to 2.17–2.26 TWh per city, with implied emissions reductions rising from about 0.14 to 0.39–0.41 Mt. CO_2 per city. This indicates that coordinated planning not only shifts the cost structure but can also translate stricter targets into auditable physical displacement and abatement benefits. To avoid feasibility gaps and abrupt cost escalation at high targets, a practical pathway is to retain the operability of annual accounting while progressively introducing deliverability checks and staged compliance requirements aligned with PCC exchange limits, private-wire transfer capability, and critical-node security constraints.

Storage requirements and evaluation should adopt a two-dimensional standard that explicitly distinguishes power and energy, with hour-by-hour feasibility as the core criterion. If evaluation focuses only on storage energy or annual throughput, it may underestimate hourly feasibility constraints, thereby amplifying cost jumps and feasibility risks as targets tighten. Accordingly, policy review, interconnection assessment, and project acceptance should elevate storage from a

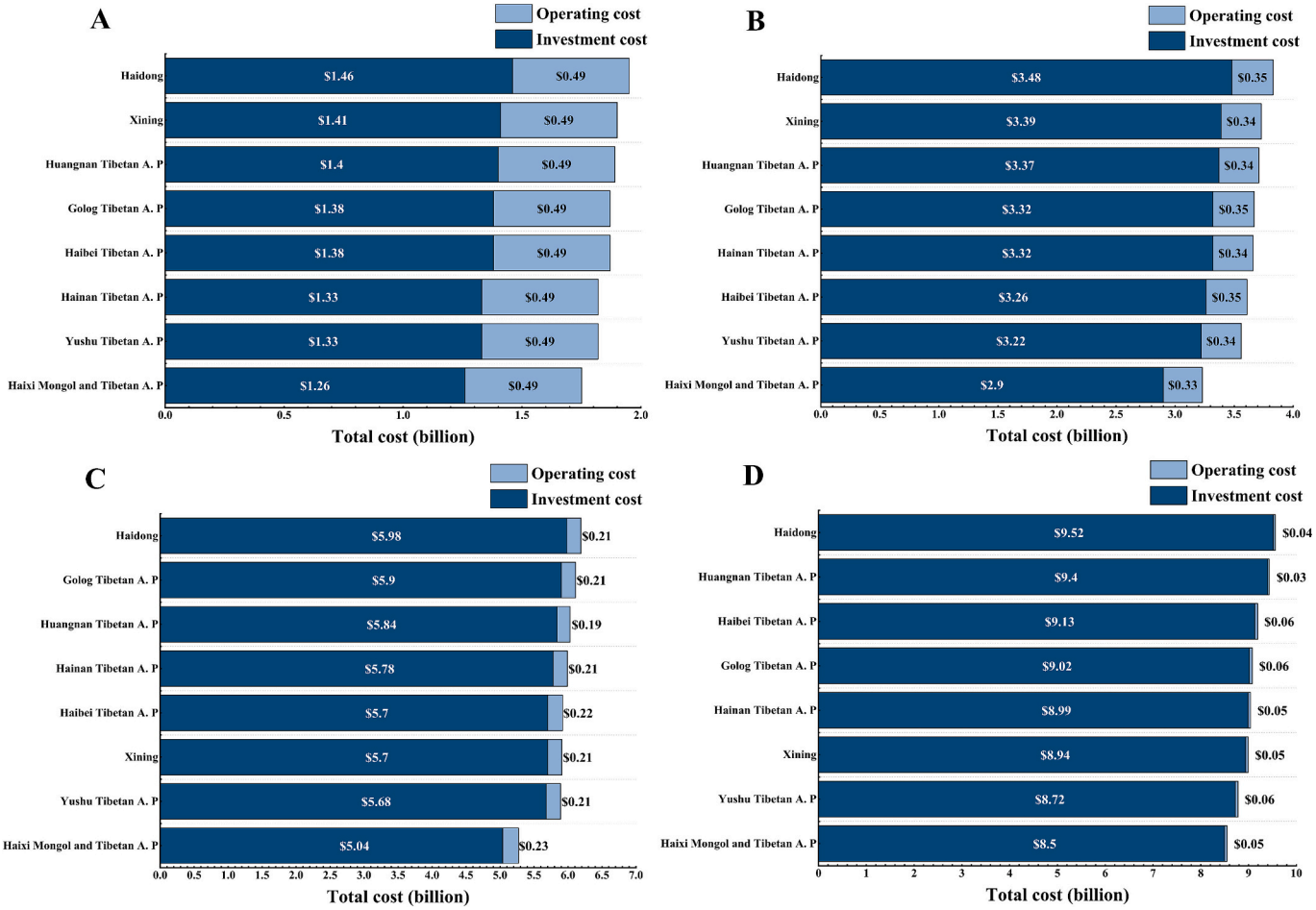


Fig. 15. Cost breakdown by city under different renewable targets. (A) 30% renewable target. (B) 50% renewable target. (C) 70% renewable target. (D) 95% renewable target.

"capacity patch" to a core asset for hourly feasibility, and explicitly recognise the different roles of power adequacy versus intertemporal energy shifting, thereby improving the auditability of compliance and operation.

From an industrial deployment perspective, DC siting should be co-optimised with cooling-climate advantages and renewable deliverability. Because total facility demand under a fixed IT scale varies with climate-driven PUE, cool, dry regions can reduce the load baseline, thereby lowering the required local wind/PV and storage build to achieve the same target, improving both economics and feasibility at high shares. An integrated screening that combines resource endowments, land developability, deliverability boundaries, and climate indicators relevant to cooling can provide a reproducible pathway for siting and configuring green power direct-connection DC projects.

5. Conclusion

Rapid growth in DCs demand heightens the tension between deep power-sector decarbonisation and the need for highly reliable electricity. We present an engineering-based planning framework for private-wire supplied DCs, modelling a DC campus as an energy hub constrained jointly by dedicated-line deliverability and PCC exchange limits, and co-optimising siting, wind/PV build-out, two-dimensional battery sizing (power and energy), line capacities, grid trading, and substation security under 8760-h constraints. The Qinghai case demonstrates implementability under realistic geography, meteorology, and network conditions, while the formulation and workflow are transferable to other resource-rich regions and private-wire projects.

In Qinghai, developable land and wind–solar potential concentrate in northern and northwestern corridors, with Haixi providing the most contiguous high-quality wind zone and a strong basis for scalable deployment, while areas such as Yushu remain fragmented under land and ecological constraints. Across tightening renewable targets, PV acts as the scalable backbone, wind enters selectively in the strongest corridors, and storage shifts from absent at low targets to feasibility-critical and more differentiated at high targets.

Quantitatively, across candidate-city cases, annual grid electricity purchase displacement increases from about 0.76 TWh at 30% to about 2.17–2.26 TWh at 95%, corresponding to implied emissions reductions of about 0.14 to 0.39–0.41 Mt. CO_2 per city. Total annual system cost rises from about 1.76–1.96 billion USD to about 8.56–9.56 billion USD. Higher targets also lead to increasingly pronounced cost divergence across candidate-city cases, highlighting the growing importance of local resource quality, corridor conditions, and system complementarity under stringent clean-electricity requirements.

A key finding is that, at high clean-electricity shares, the scarce resource is not annual energy but hour-by-hour deliverability under joint PCC and private-wire boundary constraints. Storage, therefore, becomes a feasibility-critical asset whose value depends on both power adequacy and intertemporal energy shifting. If evaluation focuses only on storage energy or annual throughput, it may underestimate hourly feasibility constraints, thereby amplifying cost jumps and feasibility risks as targets tighten. Overall, the proposed framework provides a reproducible tool for translating high-share targets into auditable, hour-by-hour, feasible plans for hyperscale DCs and for informing green power direct connection policies.

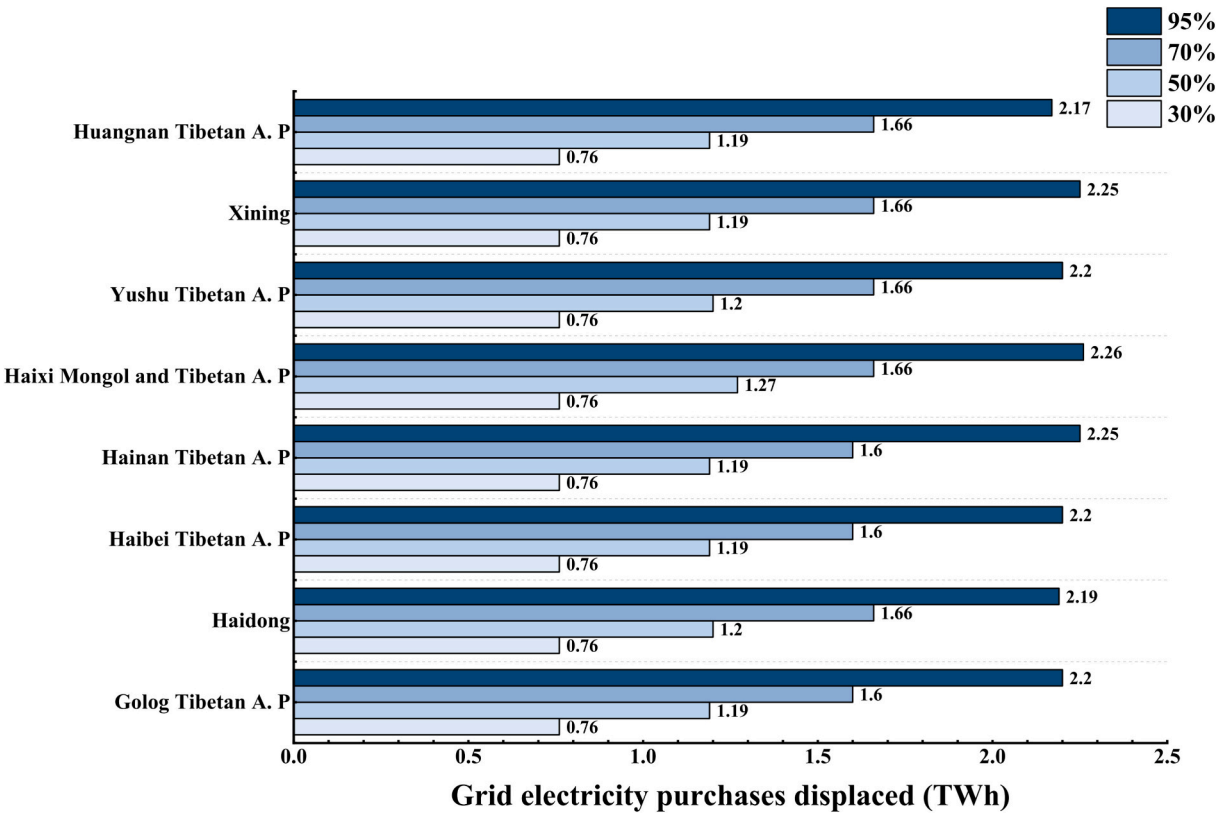


Fig. 16. Grid electricity purchases displaced (TWh) by private-wire renewables across targets.

CRediT authorship contribution statement

Feng Guo: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **João Soares:** Writing – review & editing, Validation. **Ling Ma:** Writing – review & editing, Supervision, Project administration. **Jie Song:** Resources, Funding acquisition. **Zhiyong Tian:** Resources. **Xinyu Chen:** Resources, Funding acquisition. **Zita Vale:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (Grants No. 72595830/72595835), the National Natural Science Foundation of China (72325006). F.G. acknowledges financial support from the China Scholarship Council (Grants No. 202506160024).

The Portuguese team acknowledges the research facilities and conditions provided by GECAD, partially funded by Fundação para a Ciência e a Tecnologia (FCT) (grant UID/00760/2025; doi:10.54499/UID/00760/2025).

The computation is completed in the HPC Platform of Huazhong University of Science and Technology.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2026.127841.

Appendix B. Appendix A.

Table A1
Coordinates of optimal siting results (DC, solar PV, wind).

Target	Type	Latitude	Longitude
30%	DC	35.5	98.75
30%	DC	37	102.25
30%	DC	37.75	99.75
30%	DC	37	99.25
30%	DC	38.25	92.75

(*continued on next page*)

Table A1 (*continued*)

Target	Type	Latitude	Longitude
30%	DC	35.5	94.5
30%	DC	37.25	101
30%	DC	35	101
50%	DC	35.25	98
50%	DC	37	102.25
50%	DC	37.25	99.5
50%	DC	36	99.25
50%	DC	38.75	93.25
50%	DC	35.5	94.5
50%	DC	37.25	101
50%	DC	35	101
70%	DC	35	97.75
70%	DC	37	102.25
70%	DC	37.25	99.5
70%	DC	35.75	99.25
70%	DC	37.75	92.75
70%	DC	35.5	89.75
70%	DC	36.75	101
70%	DC	35	101
95%	DC	35.25	97.75
95%	DC	36.25	101.75
95%	DC	39	99
95%	DC	36	99.25
95%	DC	37.5	95
95%	DC	35.5	89.75
95%	DC	36.75	101
95%	DC	35	101
30%	SOLAR	35.5	98.75
30%	SOLAR	36.75	102
30%	SOLAR	37.75	99.75
30%	SOLAR	37	99.25
30%	SOLAR	38.25	92.75
30%	SOLAR	35.5	94.5
30%	SOLAR	37.25	101
30%	SOLAR	35	101
50%	SOLAR	35.25	98
50%	SOLAR	36.75	102
50%	SOLAR	37.25	99.5
50%	SOLAR	36	99.25
50%	SOLAR	38.75	93.25
50%	SOLAR	35.5	94.5
50%	SOLAR	37.25	101
50%	SOLAR	35	101
70%	SOLAR	35.25	98
70%	SOLAR	36.75	102
70%	SOLAR	37.25	99.5
70%	SOLAR	35.75	99.25
70%	SOLAR	37.75	92.75
70%	SOLAR	35.5	89.75
70%	SOLAR	36.75	101
70%	SOLAR	35	101
95%	SOLAR	35.25	98
95%	SOLAR	36.25	101.75
95%	SOLAR	38.75	98.25
95%	SOLAR	36	99.25
95%	SOLAR	37.5	95
95%	SOLAR	35.5	89.75
95%	SOLAR	36.75	101
95%	SOLAR	35	101
50%	WIND	38.5	92.75
70%	WIND	38.5	92.75
95%	WIND	38.5	92.75

Data availability

We have shared our data link in the article.

References

[1] Goforth T, Nock D, Brown M, Ghosh T, Lamers P. Exploring equality and sustainability trade-offs of energy transition outcomes in the United States in 2050. Appl Energy 2024;367:123376. https://doi.org/10.1016/j.apenergy.2024.123376.

[2] Guo F, Wang J, Song Y. Research on high quality development strategy of green building: a full life cycle perspective on recycled building materials. Energ Buildings 2022;273:112406. https://doi.org/10.1016/j.enbuild.2022.112406.

[3] International Energy Agency. CO2 emissions in 2023 – Analysis [WWW document]. IEA; 2024. URL https://www.iea.org/reports/co2-emissions-in-2023 (accessed 12.16.25).

[4] Renewable Power Generation Costs in 2023. International Renewable Energy Agency. 2024.

[5] Ember. Global electricity review 2024. 2024. Ember URL https://ember-energy.org/latest-insights/global-electricity-review-2024 [accessed 12.16.25].

[6] IEA. Electricity 2025: Analysis and forecast to 2027. International Energy Agency; 2025.
[7] International Energy Agency. Energy and AI – Analysis [WWW document]. IEA; 2025. URL https://www.iea.org/reports/energy-and-ai (accessed 12.16.25).
[8] Child M, Kemfert C, Bogdanov D, Breyer C. Flexible electricity generation, grid exchange and storage for the transition to a 100% renewable energy system in europe. Renew Energy 2019;139:80–101. https://doi.org/10.1016/j.renene.2019.02.077.
[9] Guo F, Soares J, Ma L, Tian Z, Chen X, Zhang L, Vale Z. Wind and solar need storage diversity, not just capacity. The Innovation 2025;101046. https://doi.org/10.1016/j.xinn.2025.101046.
[10] Oró E, Depoorter V, Garcia A, Salom J. Energy efficiency and renewable energy integration in data centres. Strategies and modelling review. Renew Sustain Energy Rev 2015;42:429–45. https://doi.org/10.1016/j.rser.2014.10.035.
[11] Zhang H, Shao S, Xu H, Zou H, Tian C. Free cooling of data centers: a review. Renew Sustain Energy Rev 2014;35:171–82. https://doi.org/10.1016/j.rser.2014.04.017.
[12] Hintemann R, Hinterholzer S. Energy consumption of data centers worldwide how will the internet become green?. 2020.
[13] Synergy Research Group. Hyperscale data Centers hit the thousand mark; Total capacity is doubling every four years. Synergy Res. Group; 2024. URL https://www.srgresearch.com/articles/hyperscale-data-centers-hit-the-thousand-mark-total-capacity-is-doubling-every-four-years (accessed 12.16.25).
[14] Chen X, Mao H, Cheng N, Ma L, Tian Z, Luo Y, et al. Climate change impacts on global photovoltaic variability. Appl Energy 2024;374:124087. https://doi.org/10.1016/j.apenergy.2024.124087.
[15] IRENA. Renewable power generation costs in 2023. International Renewable Energy Agency, Abu Dhabi; 2024.
[16] Bird L, Lew D, Milligan M, Carlini EM, Estanqueiro A, Flynn D, et al. Wind and solar energy curtailment: a review of international experience. Renew Sustain Energy Rev 2016;65:577–86. https://doi.org/10.1016/j.rser.2016.06.082.
[17] Guo F, Ma L, Wu J, Chen K, Fang W, Broyd T. A systematic review of multi-scale digital modelling in sustainable urban design and management. Sustain Cities Soc 2025;119:106103. https://doi.org/10.1016/j.scs.2024.106103.
[18] Huo D, Zhang X, Gu W, Gao Z. Computation network and carbon neutrality: a quasi-natural experiment by externality of institutional support to energy security. Econ Anal Policy 2025;86:1427–43. https://doi.org/10.1016/j.eap.2025.05.002.
[19] Radovanovic A, Chen B, Talukdar S, Roy B, Duarte A, Shahbazi M. Power modeling for effective datacenter planning and compute management. IEEE Trans Smart Grid 2022;13:1611–21. https://doi.org/10.1109/TSG.2021.3125275.
[20] NDRC, NEA. Notice of the National Development and reform commission and the National Energy Administration on matters related to the orderly promotion of green electricity direct. 2025.
[21] Alaperä I, Honkapuro S, Paananen J. Data centers as a source of dynamic flexibility in smart girds. Appl Energy 2018;229:69–79. https://doi.org/10.1016/j.apenergy.2018.07.056.
[22] Zhou F, Gu W, Ma G. Advancements in data center cooling systems: from refrigeration to high performance cooling. Energ Buildings 2024;320:114634. https://doi.org/10.1016/j.enbuild.2024.114634.
[23] Liu Z, Liu I, Low S, Wierman A. Pricing data center demand response. SIGMETRICS Perform Eval Rev 2014;42:111–23. https://doi.org/10.1145/2637364.2592004.
[24] Radovanovic A, Koningstein R, Schneider I, Chen B, Duarte A, Roy B, et al. Carbon-aware computing for datacenters. 2021. https://doi.org/10.48550/arXiv.2106.11750.
[25] Riepin I, Brown T, Zavala VM. Spatio-temporal load shifting for truly clean computing. Adv Appl Energy 2025;17:100202. https://doi.org/10.1016/j.adapen.2024.100202.
[26] Evans Richard, Gao Jim. DeepMind AI reduces Google data Centre cooling bill by 40%. https://deepmind.google/blog/deepmind-ai-reduces-google-data-centre-cooling-bill-by-40/; 2016.
[27] Oh H, Jin W, Peng P, Winick JA, Sickinger D, Sartor D, et al. Techno-economic performance of reservoir thermal energy storage for data center cooling system. Appl Energy 2025;391:125858. https://doi.org/10.1016/j.apenergy.2025.125858.
[28] Wang T, He J, Li Y. Cooling demand response-based collaborative optimization on the supply and demand side of district cooling system under extreme heat. Sustain Cities Soc 2025;130:106550. https://doi.org/10.1016/j.scs.2025.106550.
[29] Brander M, Gillenwater M, Ascui F. Creative accounting: a critical perspective on the market-based method for reporting purchased electricity (scope 2) emissions. Energy Policy 2018;112:29–33. https://doi.org/10.1016/j.enpol.2017.09.051.
[30] Simões, F., Henriques, C., Figueiredo, N.C., Silva, P.P. da, 2025. Efficient power purchase agreement structures for meeting corporate electricity needs with solar energy. Energy 322, 135651. doi:https://doi.org/10.1016/j.energy.2025.135651.
[31] Bjørn A, Lloyd SM, Brander M, Matthews HD. Renewable energy certificates threaten the integrity of corporate science-based targets. Nat Clim Change 2022;12:539–46. https://doi.org/10.1038/s41558-022-01379-5.
[32] Bjørn A, Gebara CH, Brander M. Untangling deliverability, additionality and double counting related to renewable energy certificates for improved scope 2 emissions accounting. Environ Res Lett 2025;20:051006. https://doi.org/10.1088/1748-9326/adc941.
[33] Langer L, Bruninx K, Bjørn A, Barner L, Lavalley J, Ghadim HV, et al. Ensuring low-emission electricity purchasing requires a broader systems perspective. iScience 2025;28:112349. https://doi.org/10.1016/j.isci.2025.112349.
[34] Riepin I, Brown T. On the means, costs, and system-level impacts of 24/7 carbon-free energy procurement. Energ Strat Rev 2024;54:101488. https://doi.org/10.1016/j.esr.2024.101488.
[35] He Y, Fan J, Lin J, Li Z, Zhang J, Tang W, et al. Two-stage robust planning of data center microgrid considering batch load flexibility and multi-energy complementarity. Energy Convers Manage: X 2025;28:101266. https://doi.org/10.1016/j.ecmx.2025.101266.
[36] Rezaeimozafar M, Monaghan RFD, Barrett E, Duffy M. A review of behind-the-meter energy storage systems in smart grids. Renew Sustain Energy Rev 2022;164:112573. https://doi.org/10.1016/j.rser.2022.112573.
[37] Scottish Renewables. In: Scottish Renewables, editor. Private wire brief; 2024.
[38] European Parliament and Council. Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity. 2019.
[39] Office of Gas and Electricity Markets (Ofgem). DEWG discussion paper 2: The nature of distributed energy. [Ofgem]. 2007.
[40] Department of Climate, Energy and the Environment. Private wires policy statement. [Ireland]. 2025.
[41] Dranka GG, Ferreira P, Vaz AIF. A review of co-optimization approaches for operational and planning problems in the energy sector. Appl Energy 2021;304:117703. https://doi.org/10.1016/j.apenergy.2021.117703.
[42] Theo WL, Lim JS, Wan Alwi SR, Mohammad Rozali NE, Ho WS, Abdul-Manan Z. An MILP model for cost-optimal planning of an on-grid hybrid power system for an eco-industrial park. Energy 2016;116:1423–41.
[43] Zelaschi A, Pliotti L, Betti G, Tonno G, Sgrò D, Martelli E. An effective MILP model for the optimal design of microgrids with high-reliability requirements. Appl Energy 2025;383:125359. https://doi.org/10.1016/j.apenergy.2025.125359.
[44] Liu D, Yue J, Wang J, Espinoza M. Chapter 6: The outlook for wind and solar power. In: Embracing the future, powering growth: An energy system renewed for China. Cham: Springer Nature Switzerland; 2025. p. 177–95. https://doi.org/10.1007/978-3-031-73075-7_7.
[45] Duan M, Gao M, Jiang H, Xu R, Su B, Jiang T. Spatiotemporal change in climate variables and resources of wind, solar radiation and precipitation in Qinghai Province from 1961 to 2021. Gaoyuan Qixiang 2024;43:1614–29.
[46] He G, Kammen DM. Where, when and how much wind is available? A provincial-scale wind resource assessment for China. Energy Policy 2014;74:116–22. https://doi.org/10.1016/j.enpol.2014.07.003.
[47] Zhu R, Sun C, Yan Y. Formation mechanism and development potential of wind energy resources on the tibetan plateau. Renew Energy 2024;227:120527. https://doi.org/10.1016/j.renene.2024.120527.
[48] Ham S-W, Kim M-H, Choi B-N, Jeong J-W. Energy saving potential of various air-side economizers in a modular data center. Appl Energy 2015;138:258–75. https://doi.org/10.1016/j.apenergy.2014.10.066.
[49] Xinhua News Agency. Chinas Qinghai launches 750-kV power project connecting new energy to grid. 2024.
[50] Yang J, Huang X. The 30 m annual land cover datasets and its dynamics in China from 1985 to 2023. 2024. https://doi.org/10.5281/zenodo.12779975.
[51] UNEP-WCMC, IUCN. World database on protected areas (WDPA). 2026.
[52] Resource and Environmental Science Data Platform. China Ecological Function Zoning Dataset. 2023.
[53] UNESCO World Heritage Centre. UNESCO spatial dataset of natural and mixed World Heritage sites. 2024.
[54] FAO, IIASA, ISRIC, ESDAC. Harmonized World Soil Database v2.0. 2023.
[55] Giardini Domenico, Grünthal Gottfried, Shedlock Kaye M, Zhang Peizhen. The GSHAP global seismic hazard map. Ann Geophys 1999;42:1225–8.
[56] European Centre for Medium-Range Weather Forecasts. ERA5 hourly data on single levels from 1940 to present. 2026.
[57] Open Infrastructure Map. Open infrastructure map [WWW document]. URL 2026-03-22 (accessed 3.22.26). 2024.
[58] Ilager S, Toosi AN, Jha MR, Brandic I, Buyya R. A data-driven analysis of a cloud data Center: Statistical characterization of workload, energy and temperature, in: Proceedings of the IEEE/ACM 16th international conference on utility and cloud computing. In: Presented at the UCC '23: IEEE/ACM 16th international conference on utility and cloud computing, ACM, Taormina (Messina) Italy; 2023. p. 1–10. https://doi.org/10.1145/3603166.3632137.
[59] Masters GM. Renewable and efficient electric power systems. 2013.
[60] Wang Y, Wang R, Tanaka K, Ciais P, Penuelas J, Balkanski Y, et al. Accelerating the energy transition towards photovoltaic and wind in China. Nature 2023;619:761–7. https://doi.org/10.1038/s41586-023-06180-8.
[61] Shao X, Zhang Z, Song P, Feng Y, Wang X. A review of energy efficiency evaluation metrics for data centers. Energ Buildings 2022;271:112308. https://doi.org/10.1016/j.enbuild.2022.112308.
[62] Yuventi J, Mehdizadeh R. A critical analysis of power usage effectiveness and its use in communicating data center energy consumption. Energ Buildings 2013;64:90–4. https://doi.org/10.1016/j.enbuild.2013.04.015.

[63] Zhang Y, Li H, Wang S. Energy performance analysis of multi-chiller cooling systems for data centers concerning progressive loading throughout the lifecycle under typical climates. Build Simul 2024;17:1693–708. https://doi.org/10.1007/s12273-024-1167-9.
[64] Alduchov OA, Eskridge RE. Improved magnus form approximation of saturation vapor pressure. J Appl Meteorol 1996;35:601–9. https://doi.org/10.1175/1520-0450(1996)035%253C0601:IMFAOS%253E2.0.CO;2.
[65] Stull R. Wet-bulb temperature from relative humidity and air temperature. J Appl Meteorol Climatol 2011;50:2267–9. https://doi.org/10.1175/JAMC-D-11-0143.1.
[66] Ministry of Ecology and Environment of the People's Republic of China, National Bureau of Statistics of China. 2023 Electricity Carbon Dioxide Emission Factors (No. Announcement 2025 No. 47). 2025.
[67] Goldwind. Goldwind supports the grid connection of China's first data-center green-power direct-connection Source–Grid–Load–Storage integrated project [in Chinese]. 2025.
[68] Xia Zhipeng. In: China Energy Engineering Corporation, CEEC, editor. First-batch grid connection achieved for the Ulanqab CICC Low-carbon Computing Base source–grid–load–storage integrated project, undertaken by CEEC Tianjin electric power construction [in Chinese]; 2025.